VETENSKAPLIG DATAVISUALISERING

GEORGIOS KARAMANIS

AGENDA

Introduktion

Grunderna i datavisualisering

Grundläggande riktlinjer och vanliga fallgropar

☕ Paus

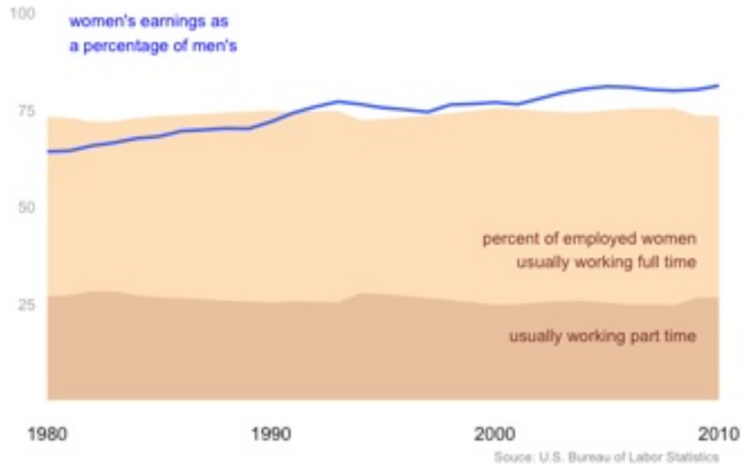
Exempel och övningar i ggplot2

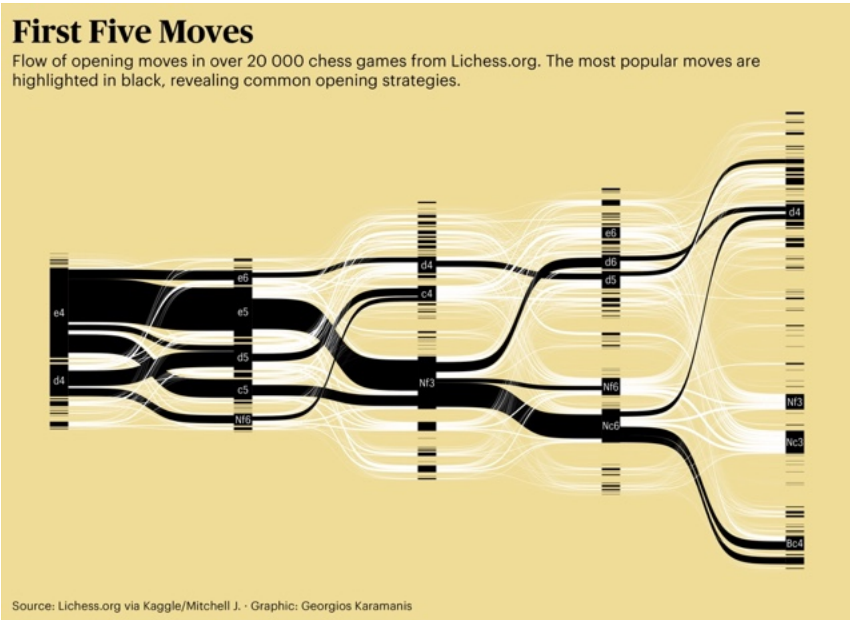
GEORGIOS KARAMANIS

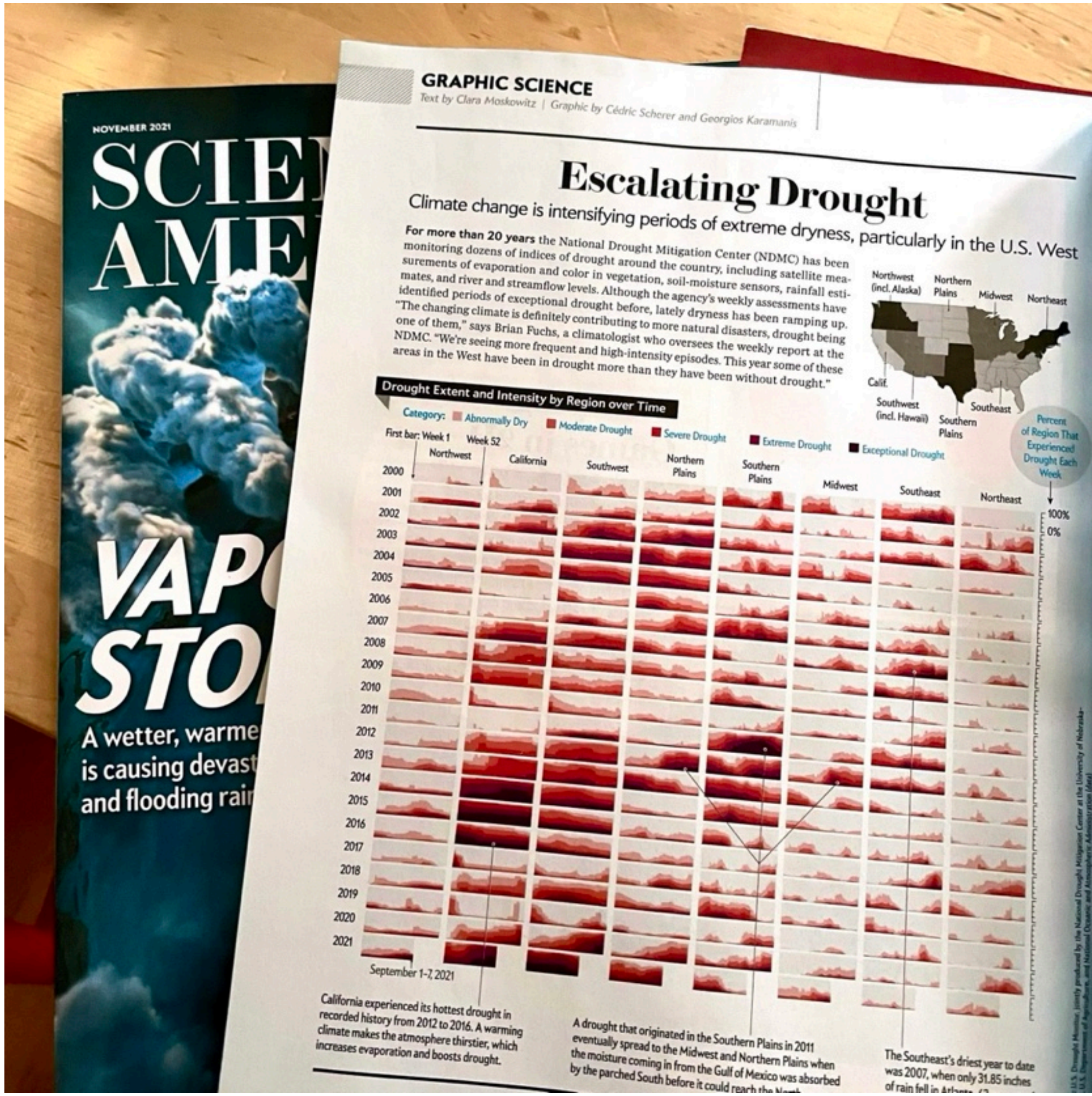
Psykiatriker, Könsidentitetsmottagningen, Akademiska sjukhuset

Doktorand, Institution för medicinska vetenskaper, Uppsala Universitet

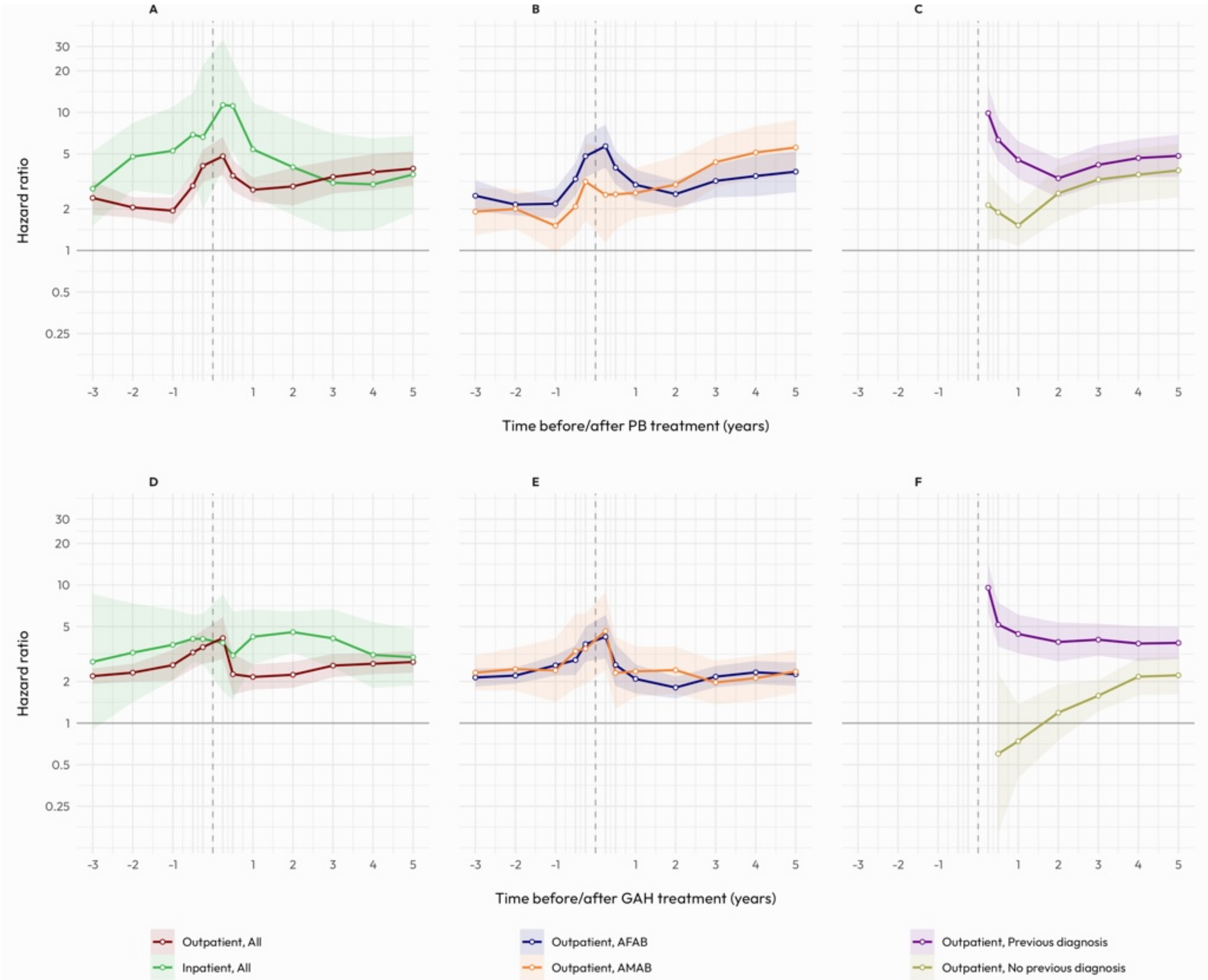
Designer inom datavisualisering, frilansare



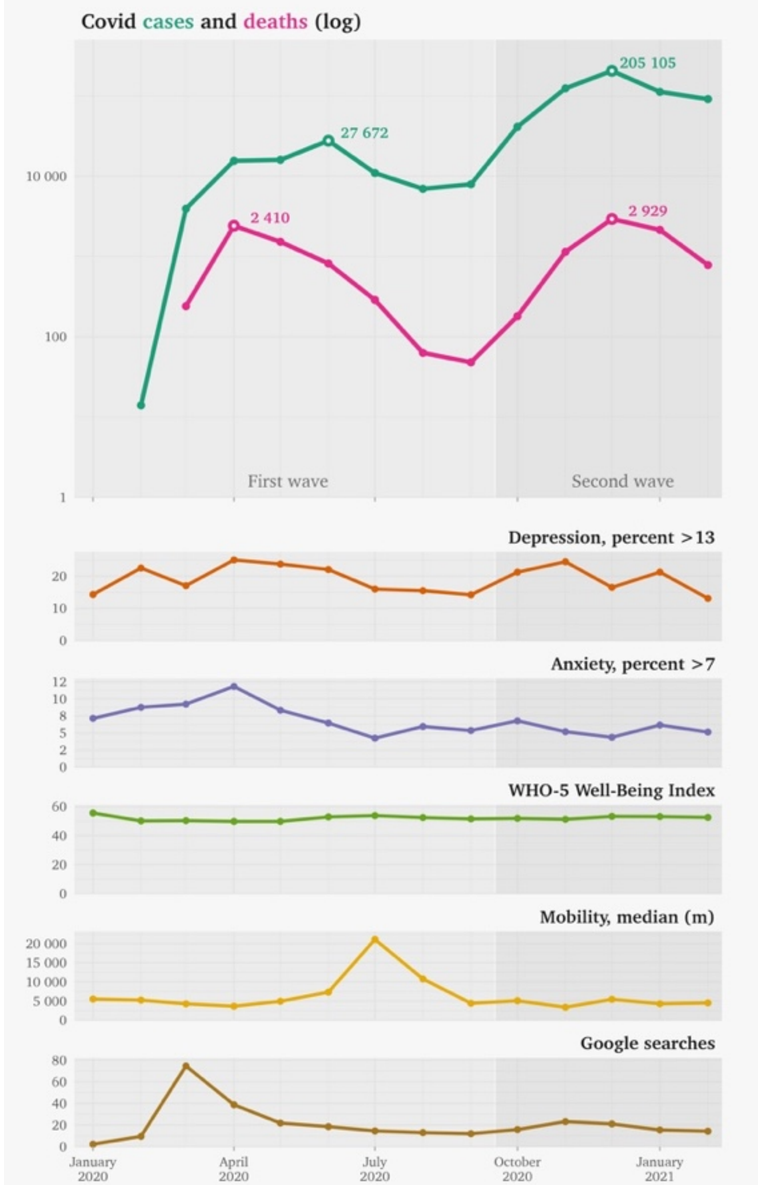




https://www.scientificamerican.com/article/climate-change-drives-escalating-drought/#



A
B
C
D
E
F
Hazard ratio
Time before/after PB treatment (years)
Time before/after GAH treatment (years)
Outpatient, All
Inpatient, All
Outpatient, AFAB
Outpatient, AMAB
Outpatient, Previous diagnosis
Outpatient, No previous diagnosis

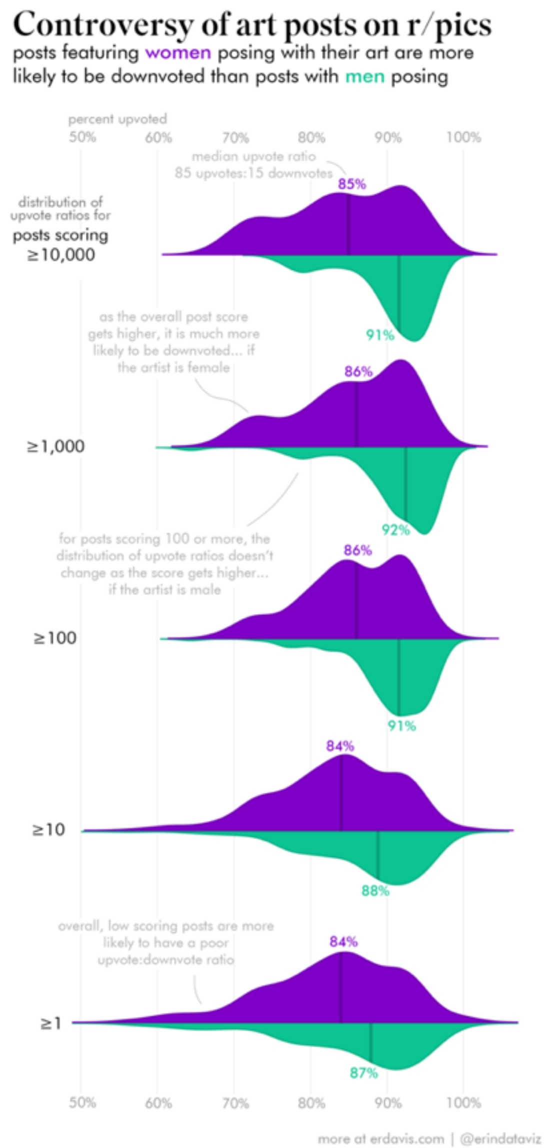


Covid cases and deaths (log)
205 105
27 672
2 410
2 929
First wave
Second wave
Depression, percent >13
Anxiety, percent >7
WHO-5 Well-Being Index
Mobility, median (m)
Google searches
January 2020
April 2020
July 2020
October 2020
January 2021

INTRODUKTION

DATAVISUALISERING

Datavisualisering blandar konst och vetenskap för att förmedla berättelser från data genom grafiska representationer

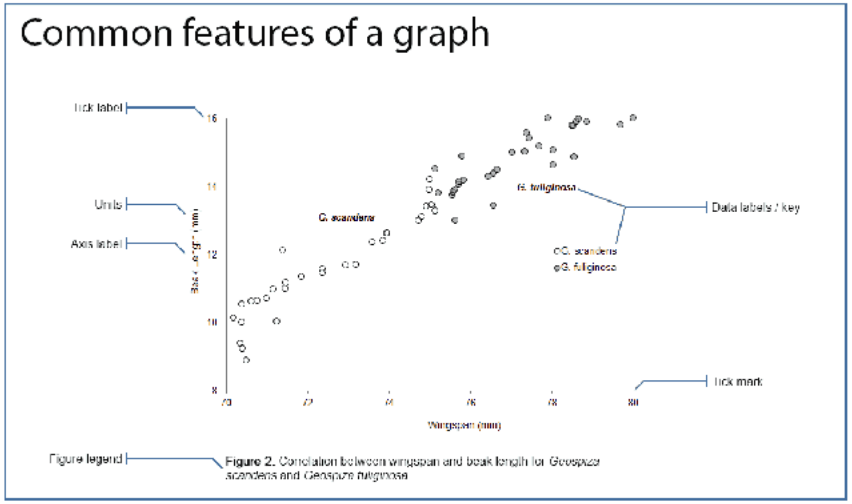


https://erdavis.com/2021/06/14/do-women-who-pose-with-their-art-on-reddit-get-more-upvotes/

SCIENTIFIC CHARTS

“Diagram i vetenskapliga publikationer är ytterst viktiga eftersom de ofta visar data som stödjer nyckelfynd”

Weissgerber TL, Milic NM, Winham SJ, Garovic VD (2015) Beyond Bar and Line Graphs: Time for a New Data Presentation Paradigm. PLoS Biol 13(4): e1002128. https://doi.org/10.1371/journal.pbio.1002128



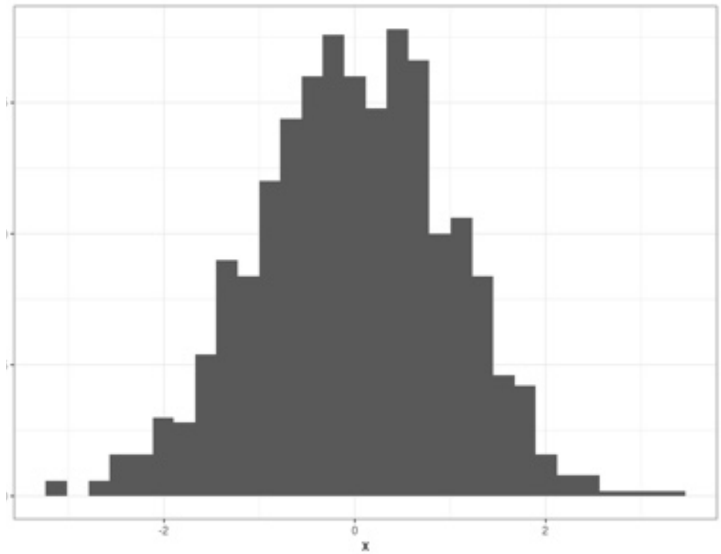
https://www.clips.edu.au/displaying-data/

VARFÖR DIAGRAM?

Lättare överblick över data

(särskilt komplexa data)

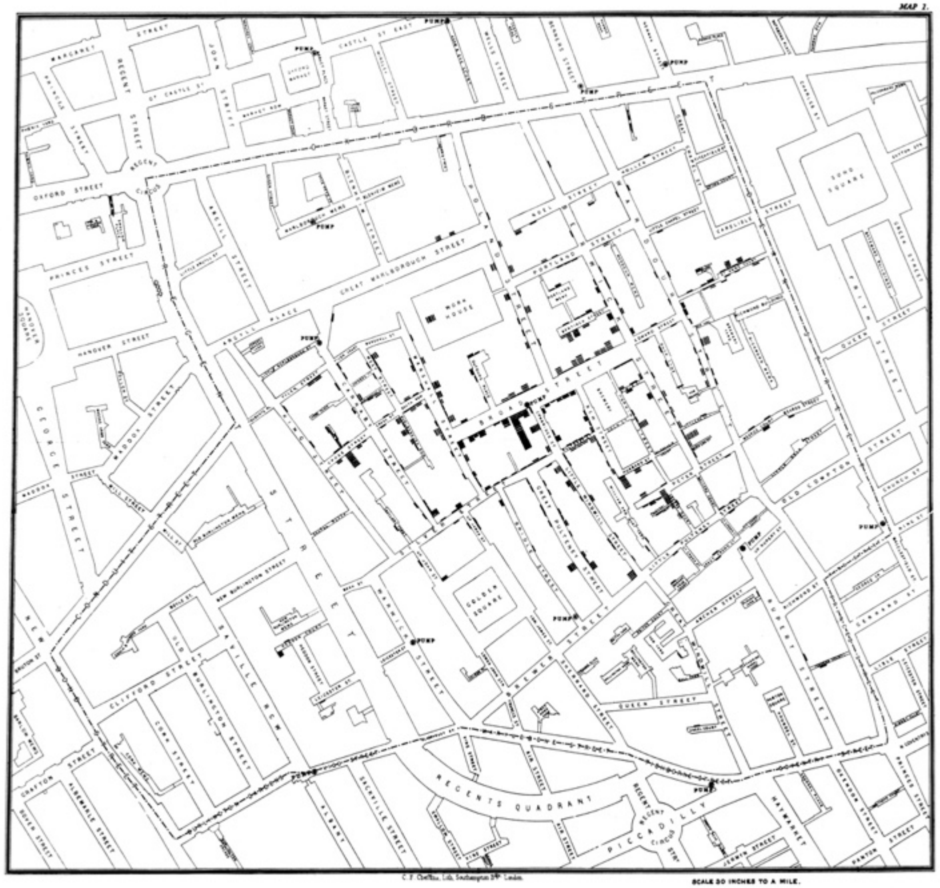
	x
1	0.376935655
2	–0.603182135
3	0.659734512
4	–1.597222895
5	–0.645683665
6	–0.929817315
7	1.401684999
8	0.520906260
9	0.369003441
10	0.573936602
11	1.304568093
12	–2.537849064
13	0.920903182
14	1.419055702
15	–0.288114527
16	1.842139177
17	0.647224489
18	1.793445537
19	–2.002214213
20	0.885130196
21	–0.369039857
22	–0.136582806



VARFÖR DIAGRAM?

Hitta nya mönster

BROAD STREET PUMP OUTBREAK, JOHN SNOW, 1854

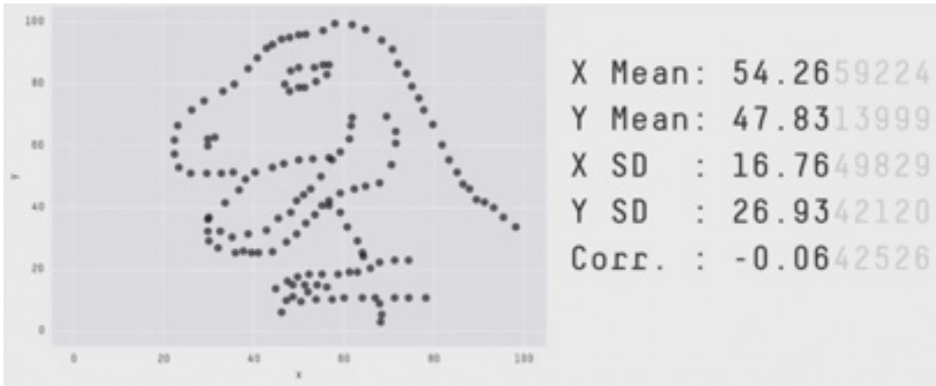


https://en.wikipedia.org/wiki/1854_Broad_Street_cholera_outbreak

VARFÖR DIAGRAM?

Enklare för jämförelse av data och deras statistiska egenskaper

THE DATASAURUS DOZEN

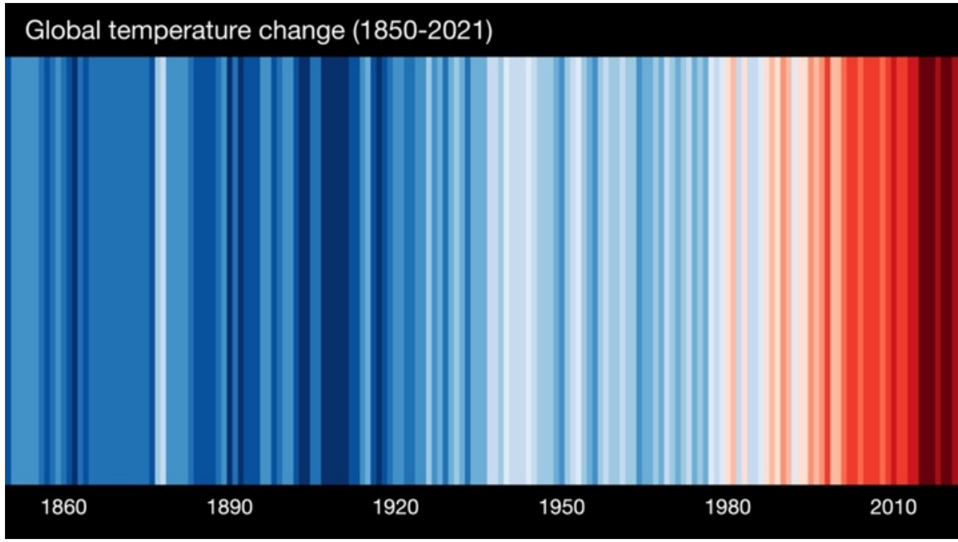


https://www.autodesk.com/research/publications/same-stats-different-graphs

VARFÖR DIAGRAM?

För att bättre kommunicera budskapet

WARMING STRIPES, ED HAWKINGS, 2018

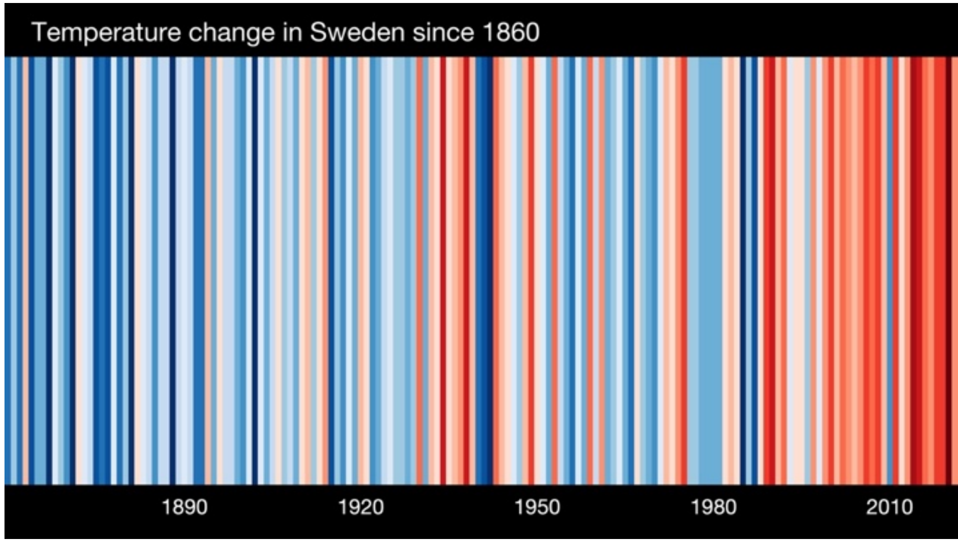


https://www.climate-lab-book.ac.uk/2018/warming-stripes/

VARFÖR DIAGRAM?

För att bättre kommunicera budskapet

WARMING STRIPES, ED HAWKINGS, 2018

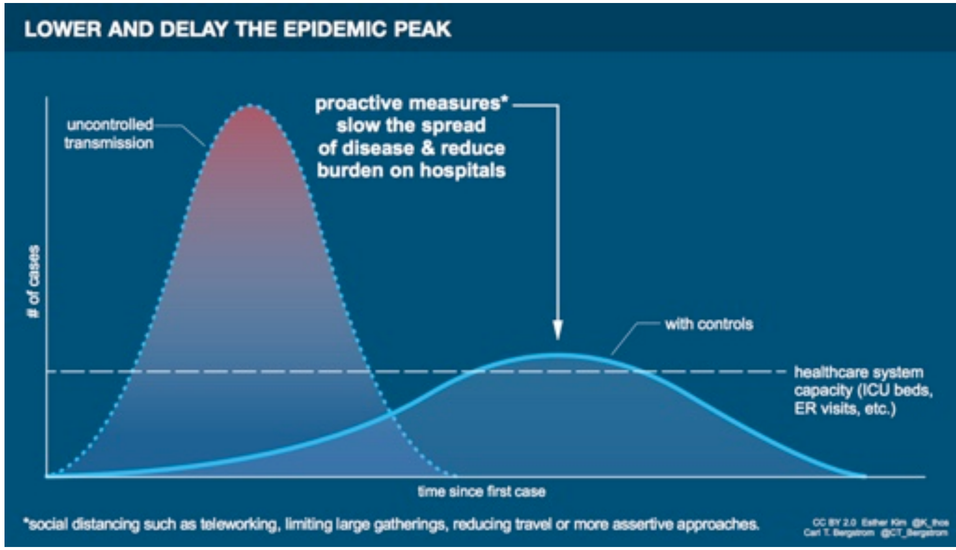


https://showyourstripes.info/l/europe/sweden/all

VARFÖR DIAGRAM?

För att bättre kommunicera budskapet

ESTHER KIM & CARL T. BERGSTROM, 2019



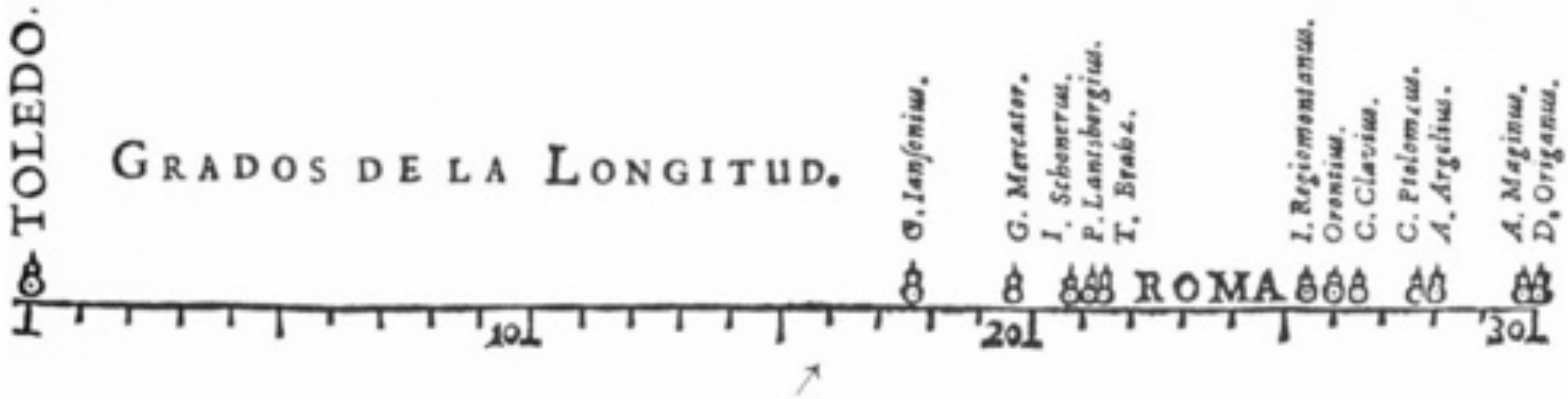
http://ctbergstrom.com/covid19.html


we're sick of this SHIT
1850
2018


MUN 0 - 0 REA 28:04

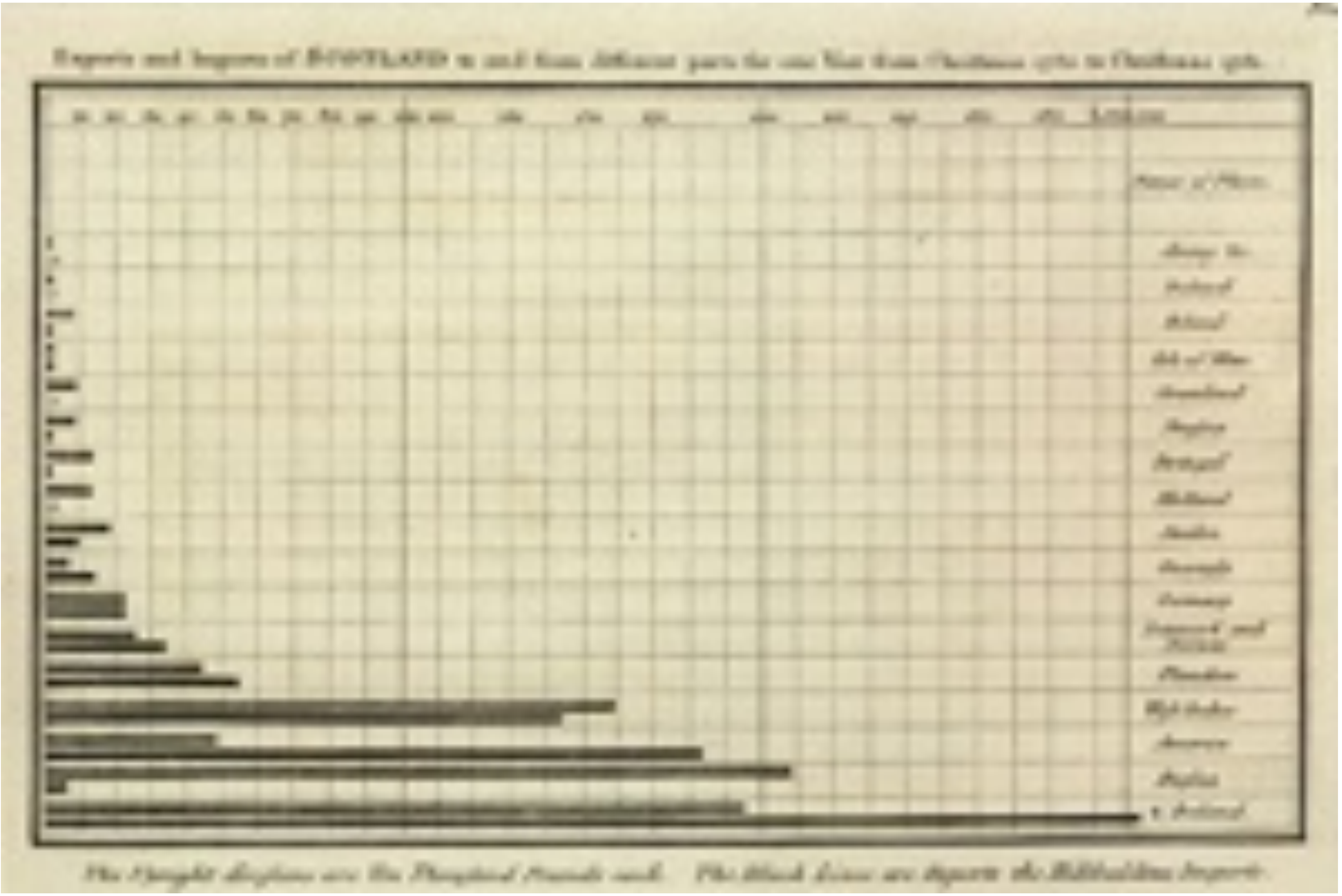
VETENSKAPLIGA DIAGRAM HAR EN LÅNG HISTORIA

MICHAEL FLORENT VAN LANGREN, 1644



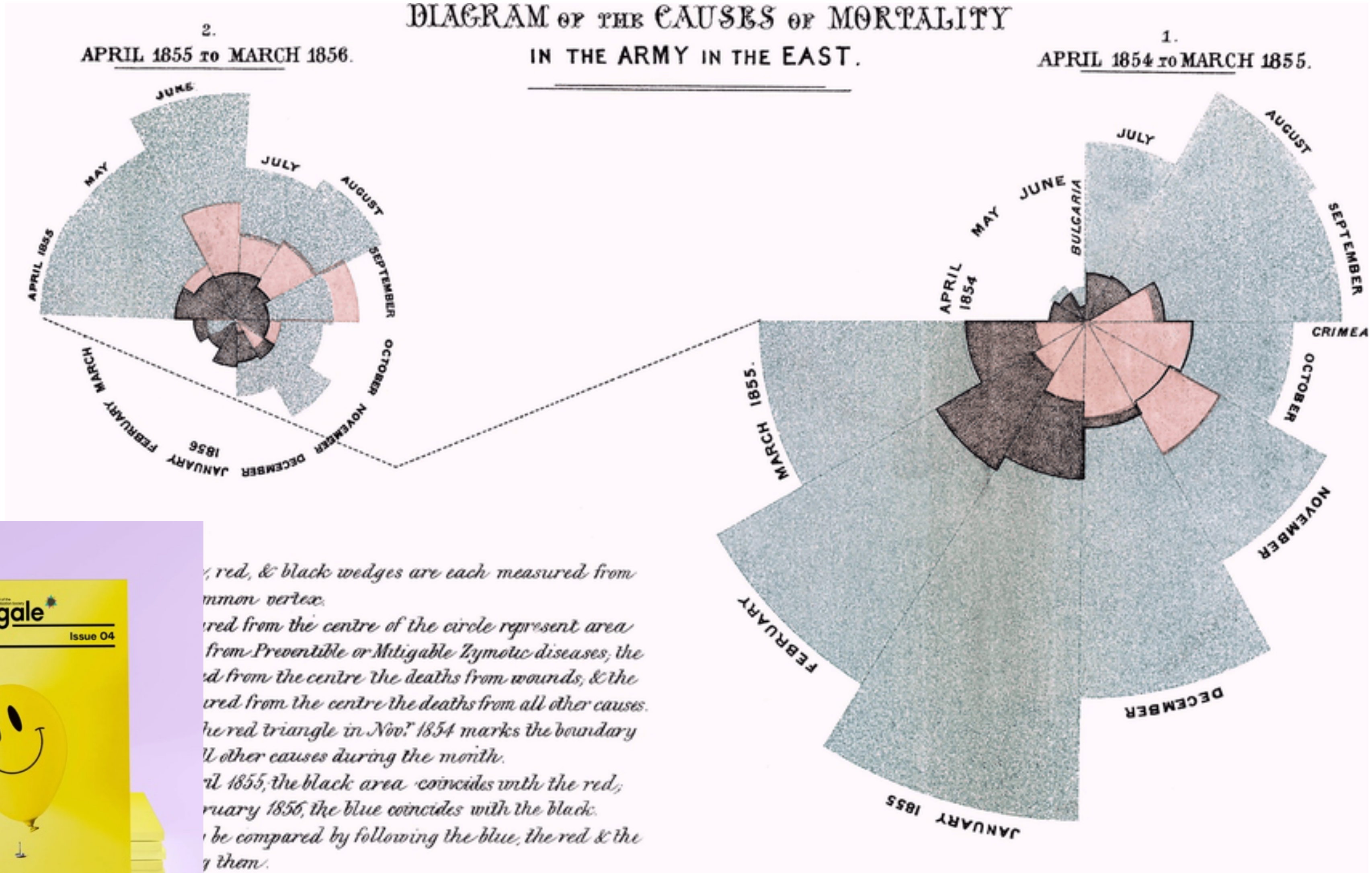
https://historyofinformation.com/detail.php?id=3415

WILLIAM PLAYFAIR, 1786



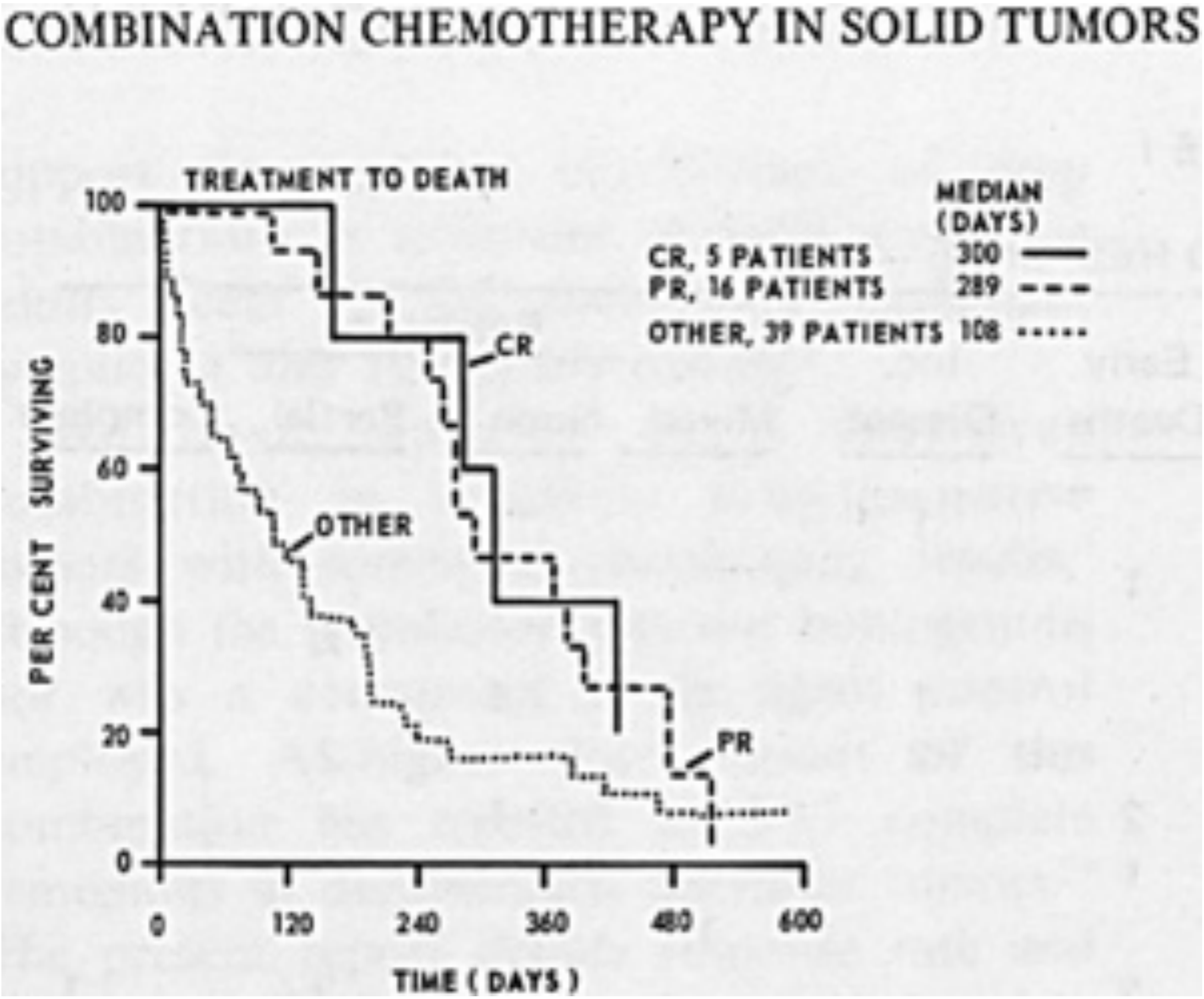
https://data-xtractor.com/blog/data-visualization/chart-history-who-invented-this-chart-type/

DIAGRAM OF THE CAUSES OF MORTALITY IN THE ARMY IN THE EAST, FLORENCE NIGHTINGALE, 1858



https://en.wikipedia.org/wiki/Florence_Nightingale

EDWARD GEHAN, 1971

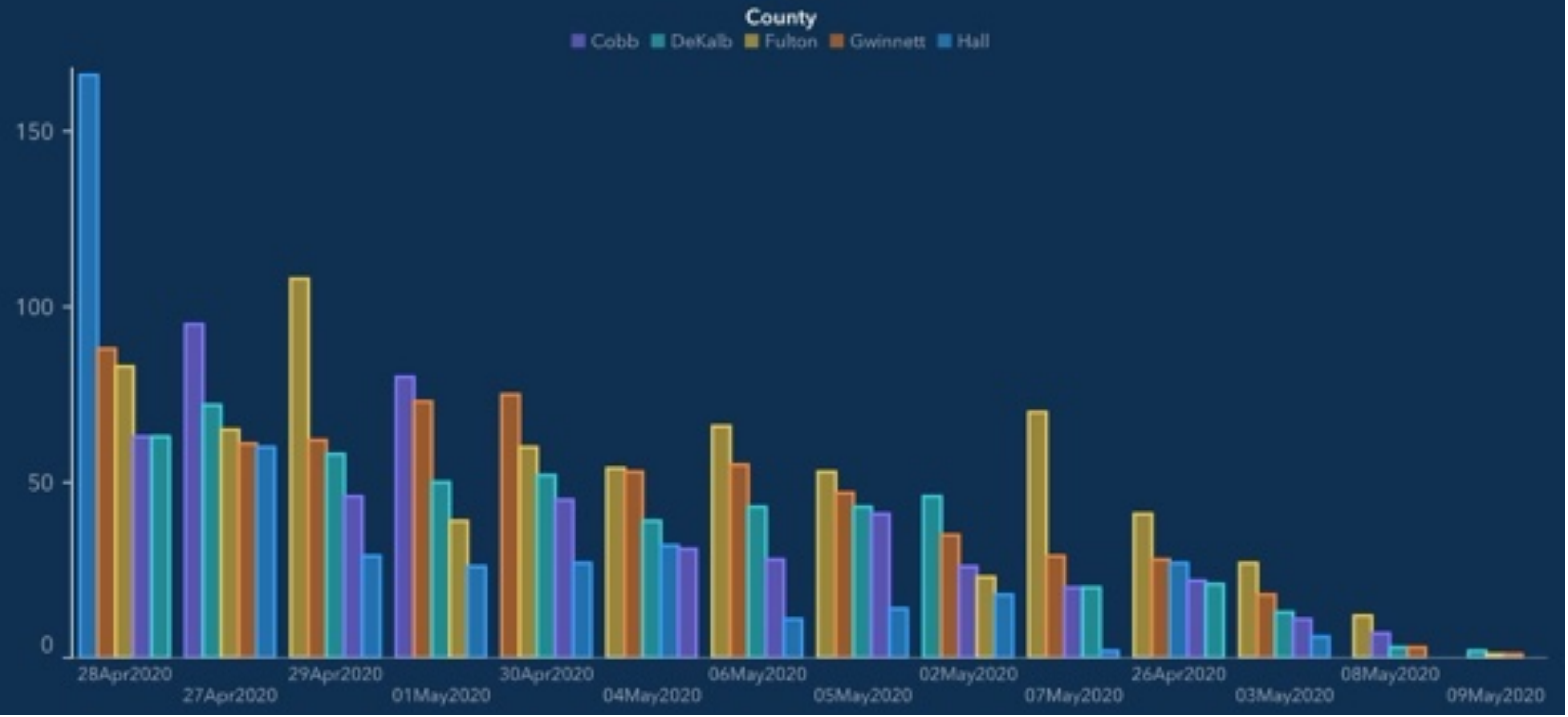


https://www.tandfonline.com/doi/full/10.1080/17498430.2018.1450055#d1e341

DÅLIGA DIAGRAM ÄR SVÅRA ATT LÄSA, KAN SKADA VÅRT BUDSKAP OCH ÄVEN VÅR TROVÄRDIGHET

Top 5 Counties with the Greatest Number of Confirmed COVID-19 Cases

The chart below represents the most impacted counties over the past 15 days and the number of cases over time. The table below also represents the number of deaths and hospitalizations in each of those impacted counties.



TECHNOLOGY FEATURE | 09 December 2024

Bad bar charts distort data — and pervade biology

Scientific 'shorthand' could be misrepresenting findings in published papers.

By Amanda Heidt

Bar charts are ubiquitous in the life-science literature, yet a study suggests that they're often used in ways that can misrepresent research findings. The preprint[1], which was first posted on bioRxiv in September and has yet to be peer reviewed, found that in a collection of nearly 3,400 papers from 2023 that included at least one bar chart, almost one-third distorted the data in some way, highlighting a need for increased data literacy among scientists and for a

Beyond Bar and Line Graphs: Time for a New Data Presentation Paradigm

Tracey L Weissgerber [1,*], Natasa M Milic [1,2], Stacey J Winham [3], Vesna D Garovic [1]

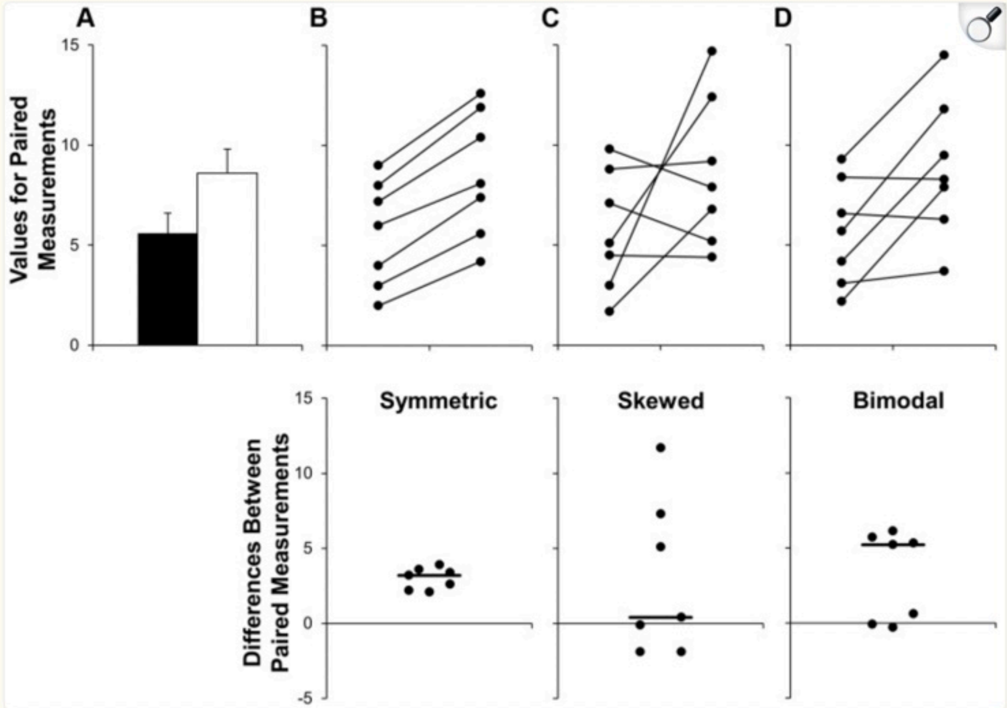
▸ Author information ▸ Article notes ▸ Copyright and License information

PMCID: PMC4406565 PMID: 25901488

Abstract

Figures in scientific publications are critically important because they often show the data supporting key findings. Our systematic review of research articles published in top physiology journals ($n = 703$) suggests that, as scientists, we urgently need to change our practices for presenting continuous data in small sample size studies. Papers rarely included scatterplots, box plots, and histograms that allow readers to critically evaluate continuous data. Most papers presented continuous data in bar and line graphs. This is problematic, as many different data distributions can lead to the same bar or line graph. The full data may suggest different conclusions from the summary statistics. We recommend training investigators in data presentation, encouraging a more complete presentation of data, and changing journal editorial policies. Investigators can quickly make univariate scatterplots for small sample size studies using our Excel templates.

Fig 2. Additional problems with using bar graphs to show paired data.



GRUNDERNA I DATAVISUALISERING

VISUELLA DESIGNELEMENT

position

längd

vinkel

riktning

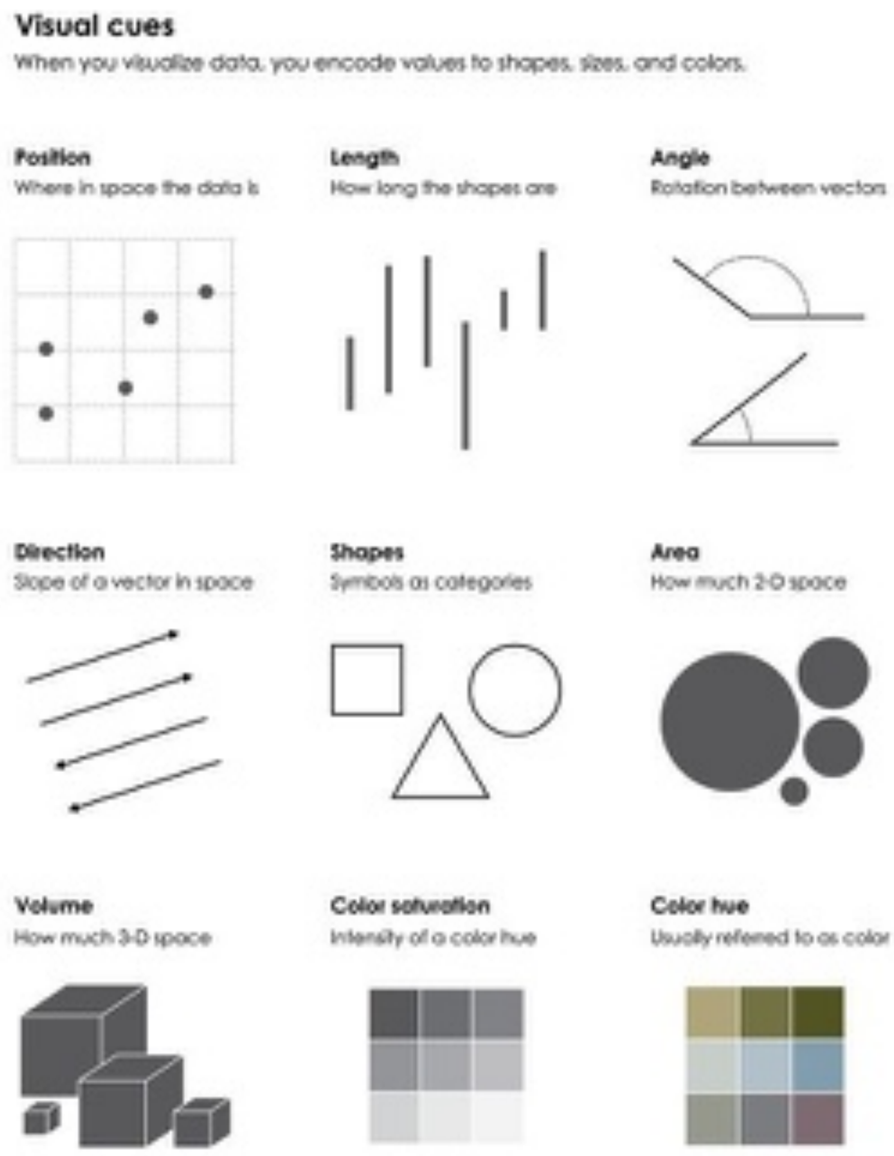
form

area

volym

färgmättnad/lätthet

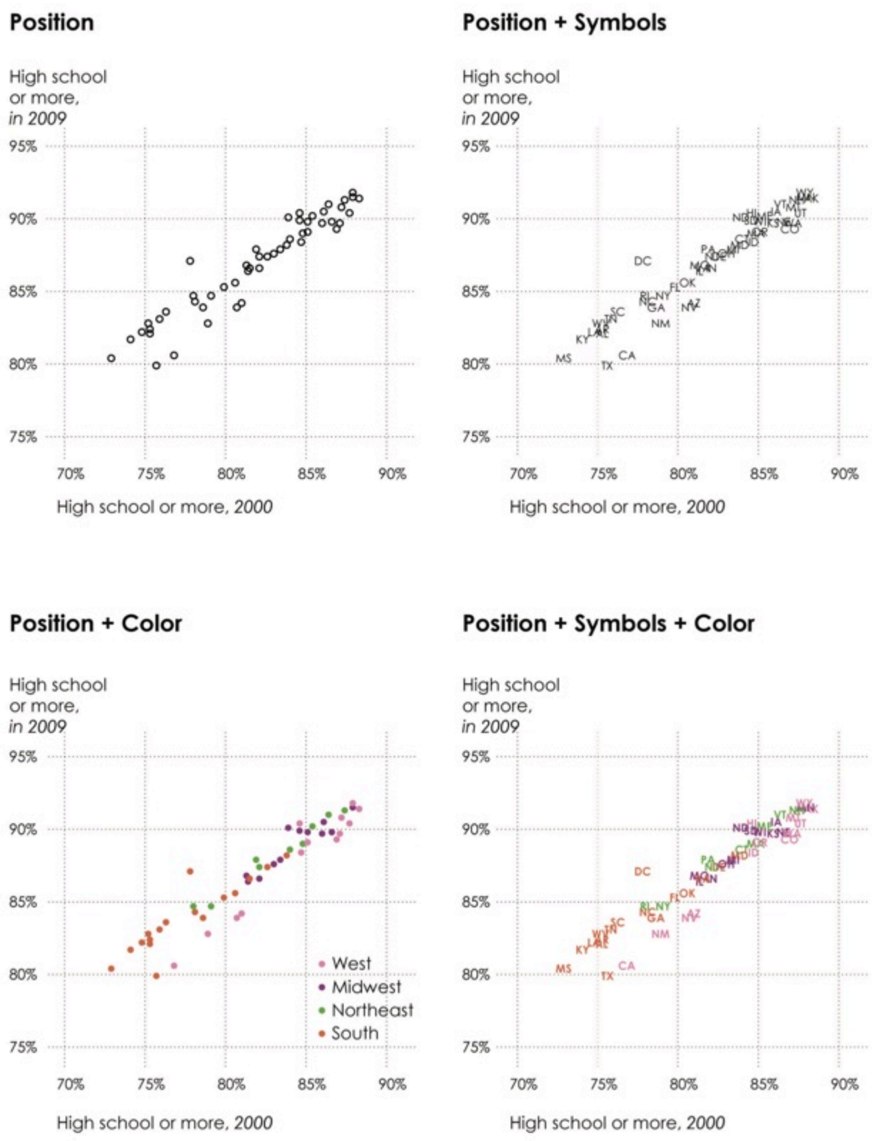
kulör (färg)



https://flowingdata.com/data-points/DataPoints-Ch3.pdf

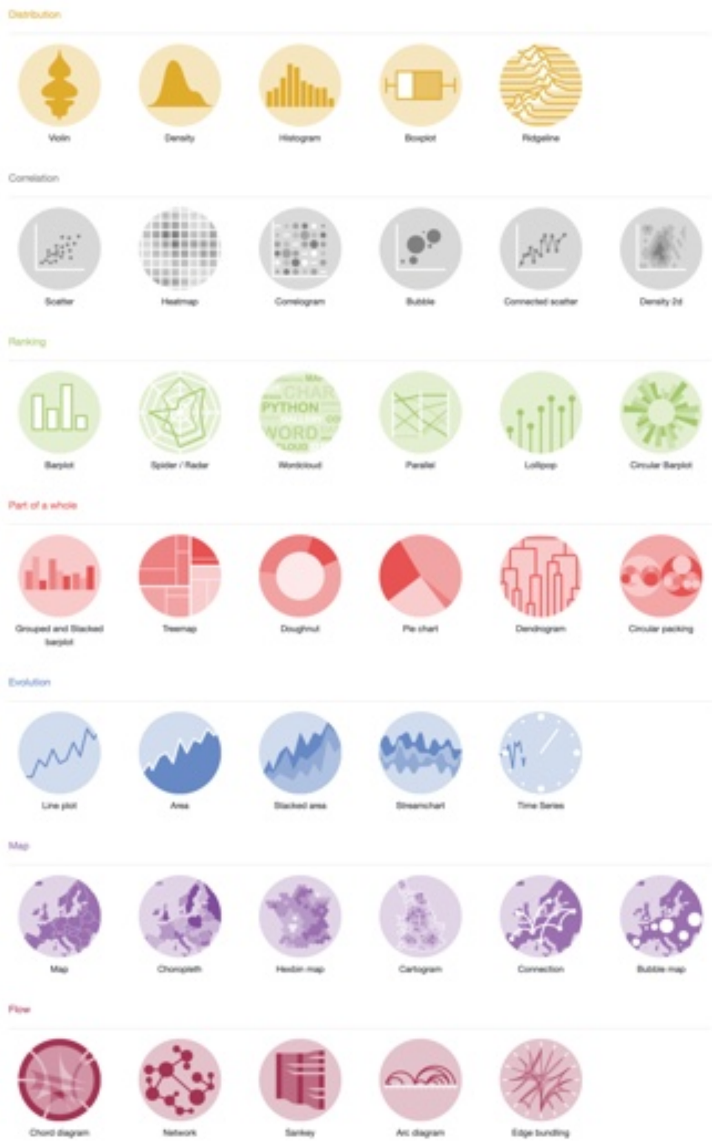
VISUELLA DESIGNELEMENT

kombinationer



https://flowingdata.com/data-points/DataPoints-Ch3.pdf

HUVUDTYPER AV DIAGRAM



https://r-graph-gallery.com

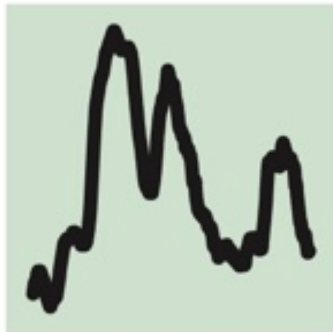
PUNKTDIAGRAM

Scatterplot

The standard way to show the relationship between two continuous variables, each of which has its own axis.

LINJEDIAGRAM

Line



The standard way to show a changing time series. If data are irregular, consider markers to represent data points.

HISTOGRAM

Histogram

The standard way to show a statistical distribution - keep the gaps between columns small to highlight the 'shape' of the data.

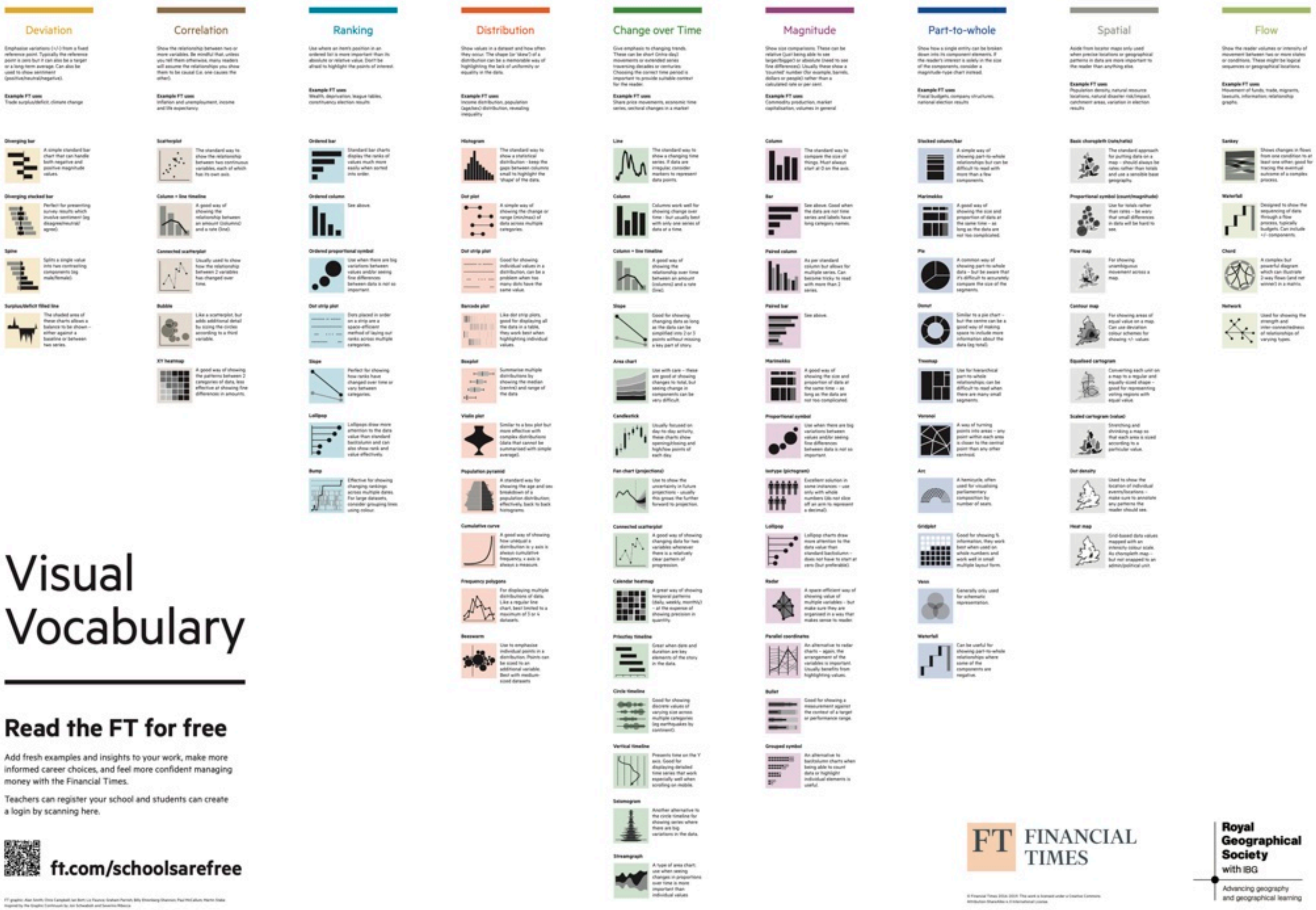
STAPELDIAGRAM

Column

The standard way to compare the size of things. Must always start at 0 on the axis.

Bar

See above. Good when the data are not time series and labels have long category names.



https://ft-interactive.github.io/visual-vocabulary/

GRUNDLÄGGANDE RIKTLINJER OCH VANLIGA FALLGROPAR

Ten guidelines for effective data visualization in scientific publications

ELSEVIER

Environmental Modelling & Software
Volume 26, Issue 6, June 2011, Pages 822-827

Short communication

Ten guidelines for effective data visualization in scientific publications

Christa Kelleher, Thorsten Wagener

Show more

Share Cite

https://doi.org/10.1016/j.envsoft.2010.12.006

Get rights and content

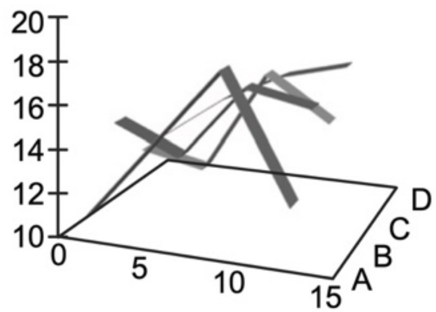
Abstract

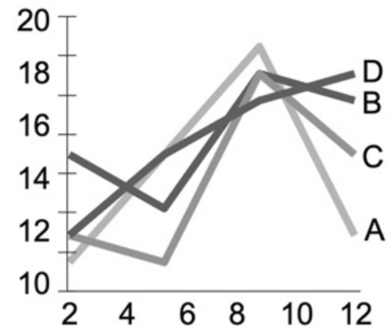
Our ability to visualize scientific data has evolved significantly over the last 40 years. However, this advancement does not necessarily alleviate many common pitfalls in visualization for scientific journals, which can inhibit the ability of readers to effectively understand the information presented. To address this issue within the context of visualizing environmental data, we list ten guidelines for effective data visualization in scientific publications. These guidelines support the primary objective of data visualization, i.e. to effectively convey information. We believe that this small set of guidelines based on a review of key visualization literature can help researchers improve the communication of their results using effective visualization. Enhancement of environmental data visualization will further improve research presentation and communication within and across disciplines.

Christa Kelleher, Thorsten Wagener, Ten guidelines for effective data visualization in scientific publications, Environmental Modelling & Software, Volume 26, Issue 6, 2011, Pages 822-827, ISSN 1364-8152, https://doi.org/10.1016/j.envsoft.2010.12.006.

RIKTLINJE 1

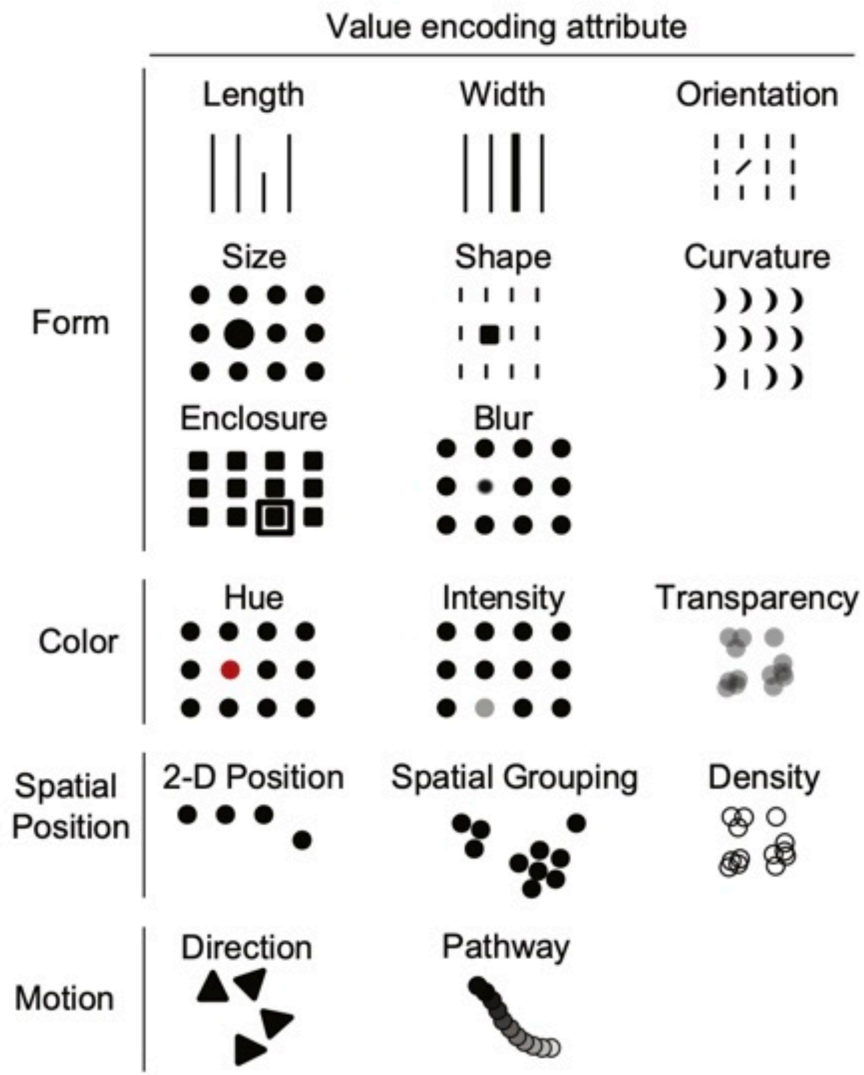
Skapa den enklaste grafen som förmedlar informationen du vill förmedla





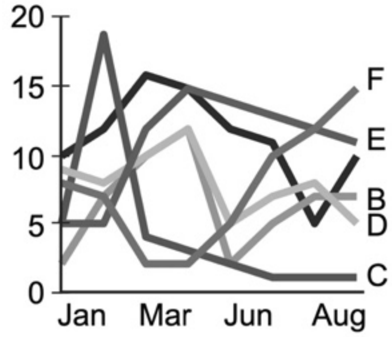
RIKTLINJE 2

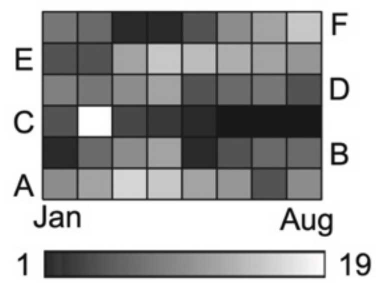
Tänk på vilket designelement ska användas



RIKTLINJE 3

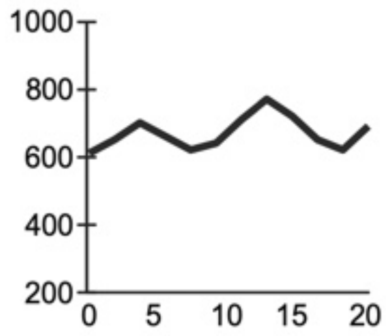
Visualisera mönster eller detaljer beroende på syftet med diagrammet

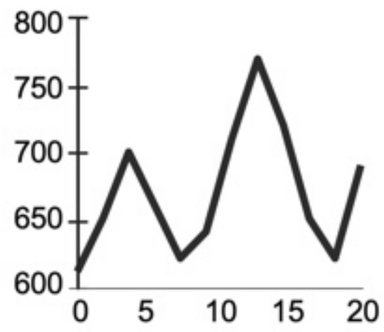


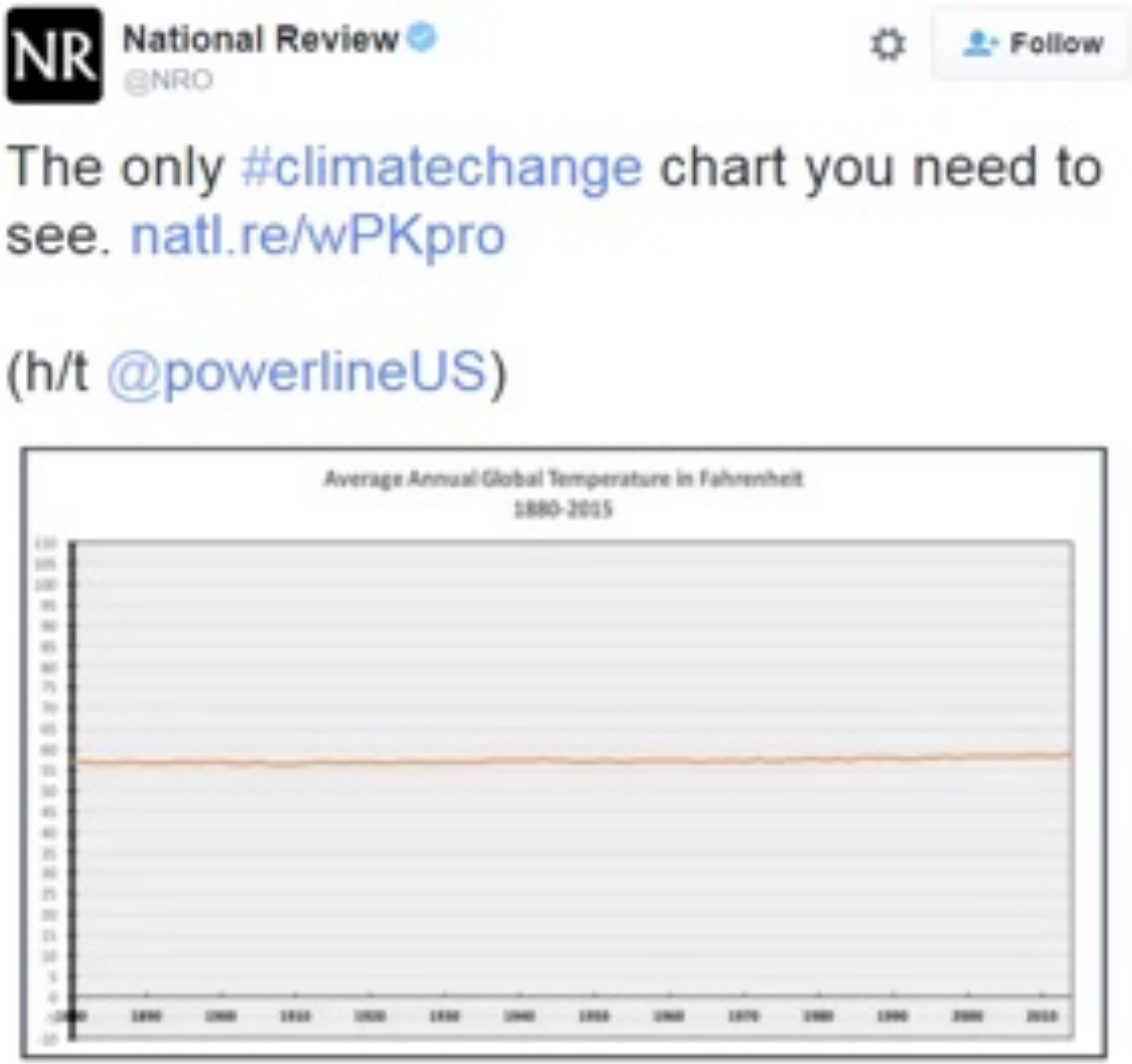


RIKTLINJE 4

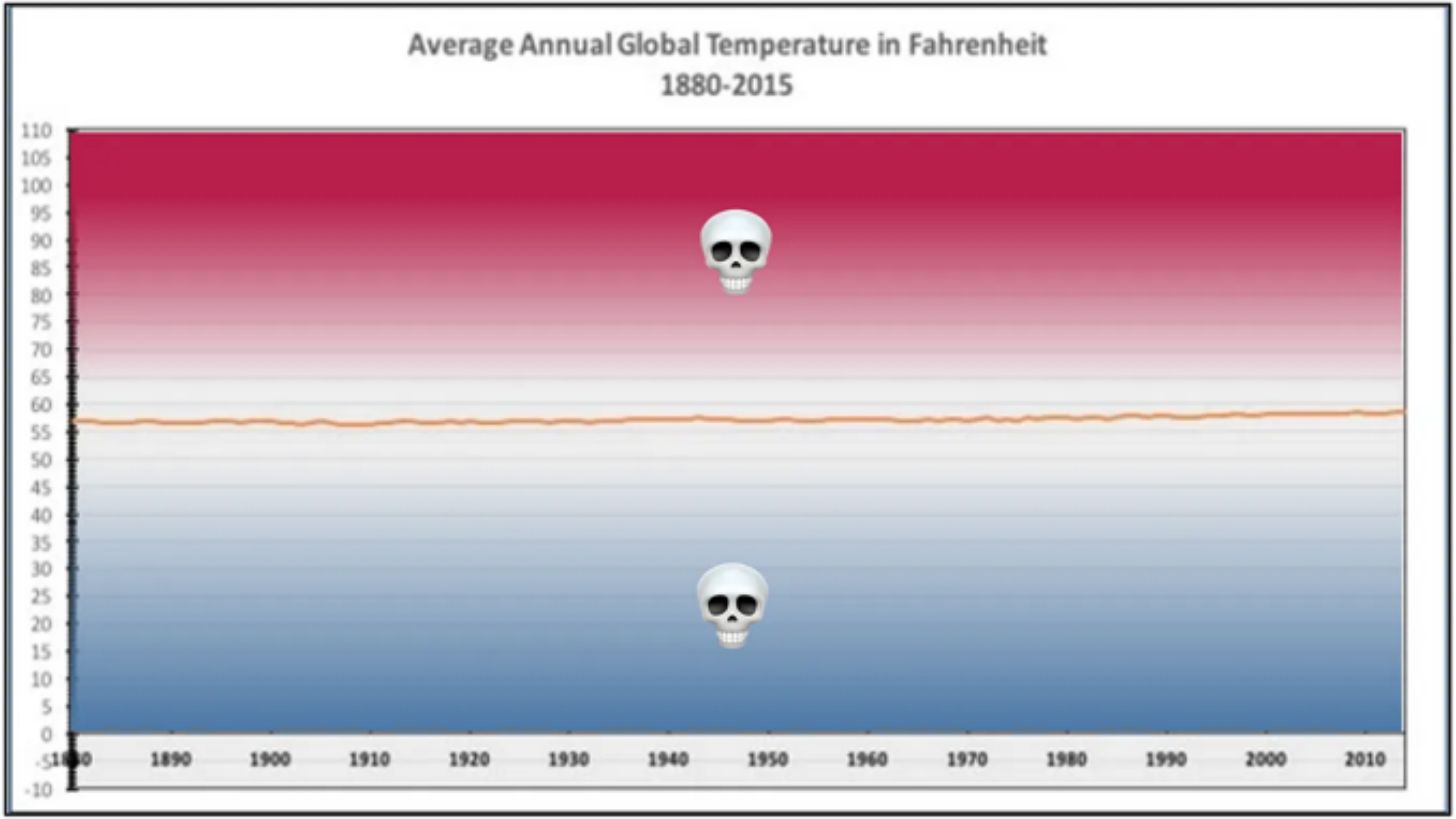
Välj meningsfulla axelskalor







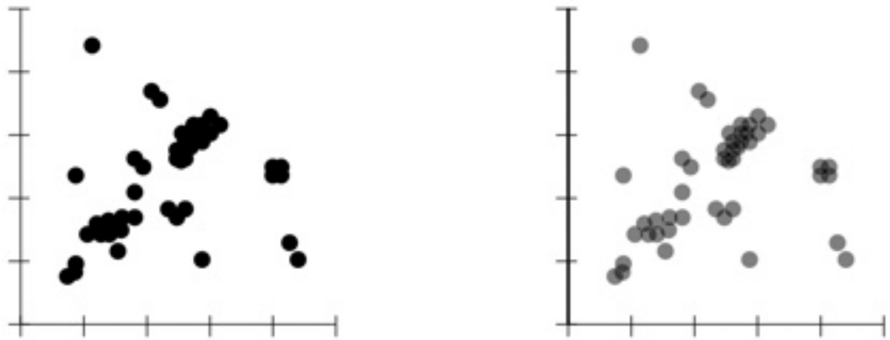
https://mcorrell.medium.com/an-inconvenient-graph-addc218579e9



https://mcorrell.medium.com/an-inconvenient-graph-addc218579e9

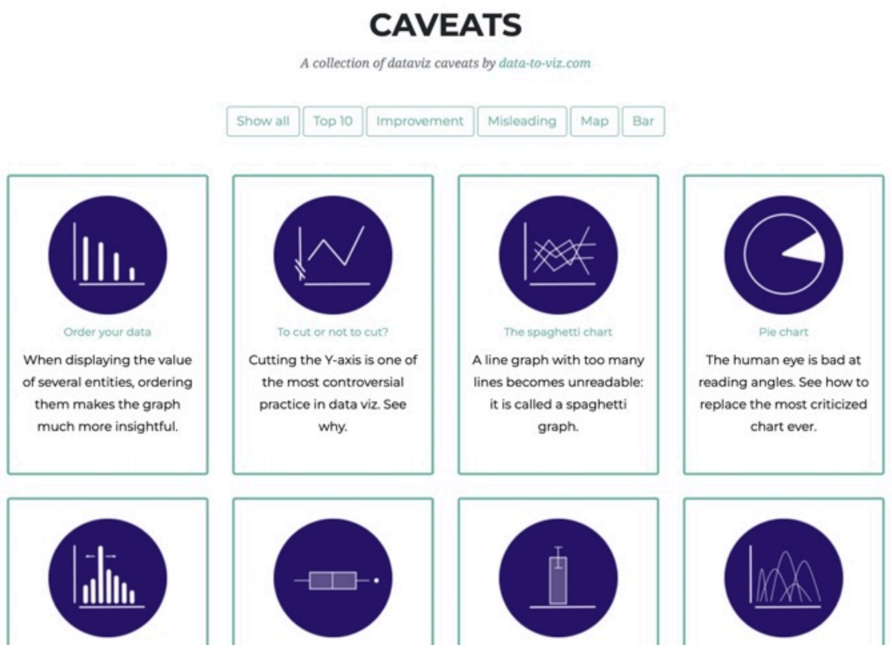
RIKTLINJE 6

Rita överlappande punkter på ett sätt så att densitetsskillnader blir uppenbara i punktdiagram



VANLIGA FALLGROPAR

OCH ALTERNATIV



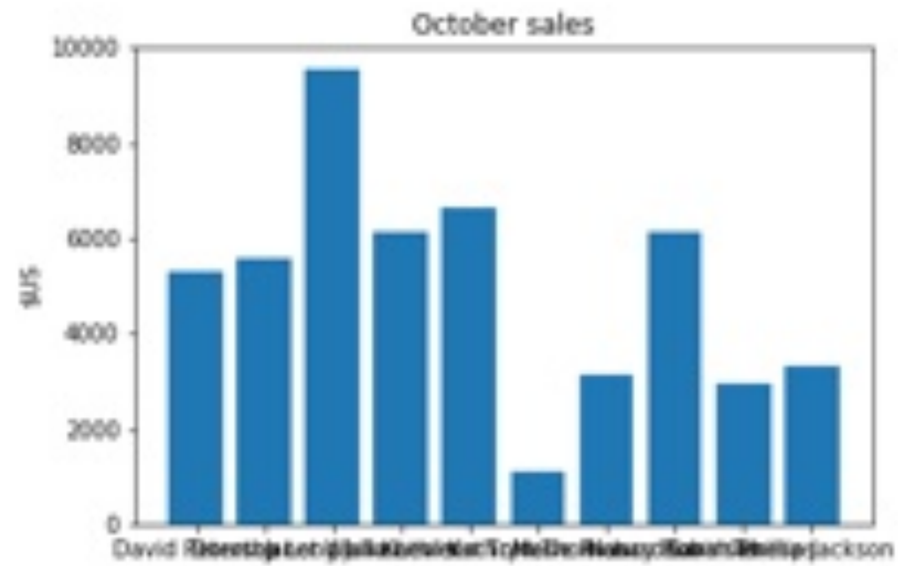
Mer på

https://www.data-to-viz.com/caveats.html

VERTIKAL AXELTEXT

Vanligt problem

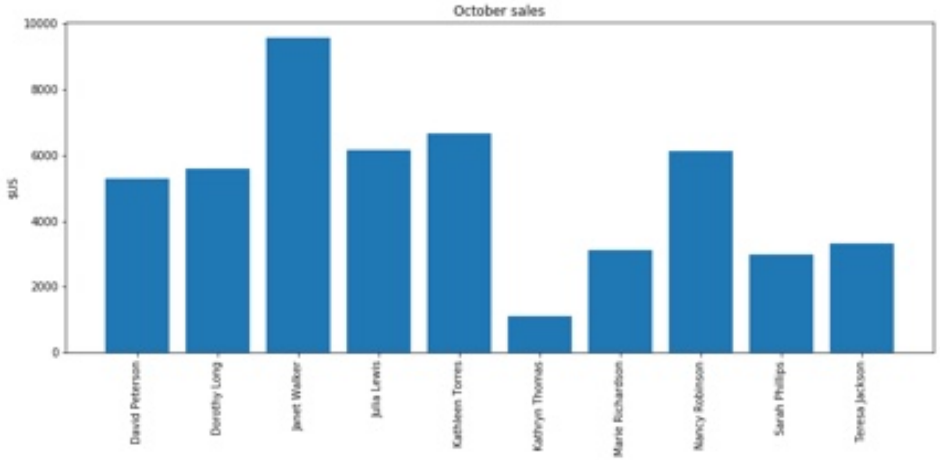
Svårläst



https://gorelik.net/2017/11/23/how-to-make-a-graph-less-readable-rotate-the-text-labels/

VERTIKAL AXELTEXT

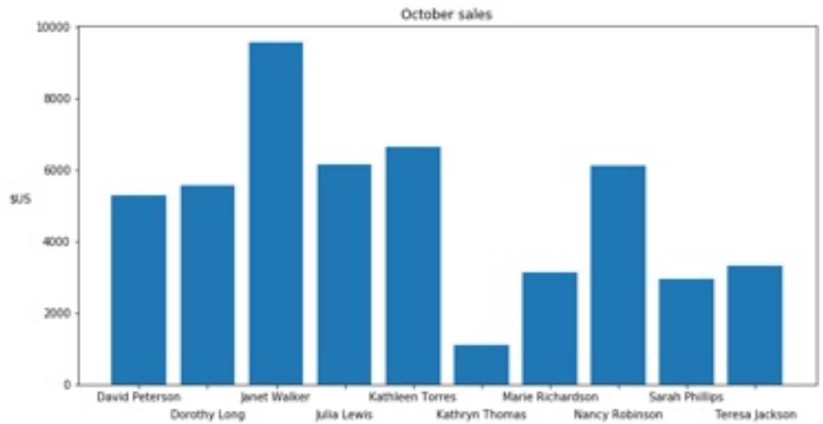
Fortfarande svårläst



https://gorelik.net/2017/11/23/how-to-make-a-graph-less-readable-rotate-the-text-labels/

VERTIKAL AXELTEXT

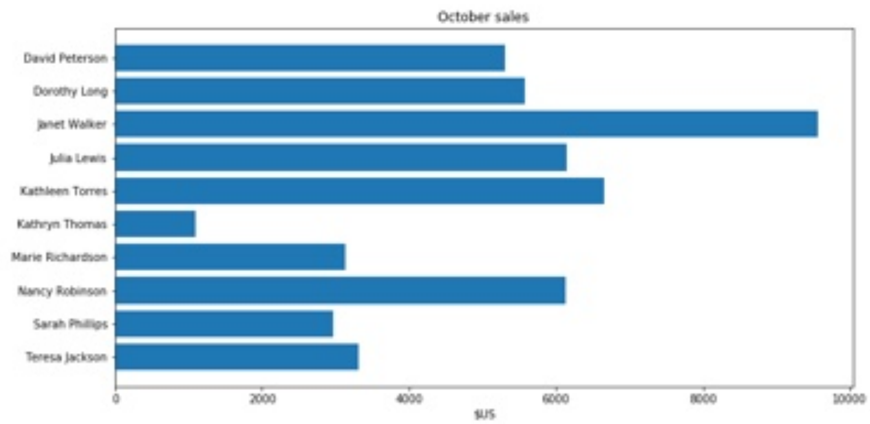
Överväg att flytta axeletiketterna



https://gorelik.net/2017/11/23/how-to-make-a-graph-less-readable-rotate-the-text-labels/

VERTIKAL AXELTEXT

Rotera staplarna

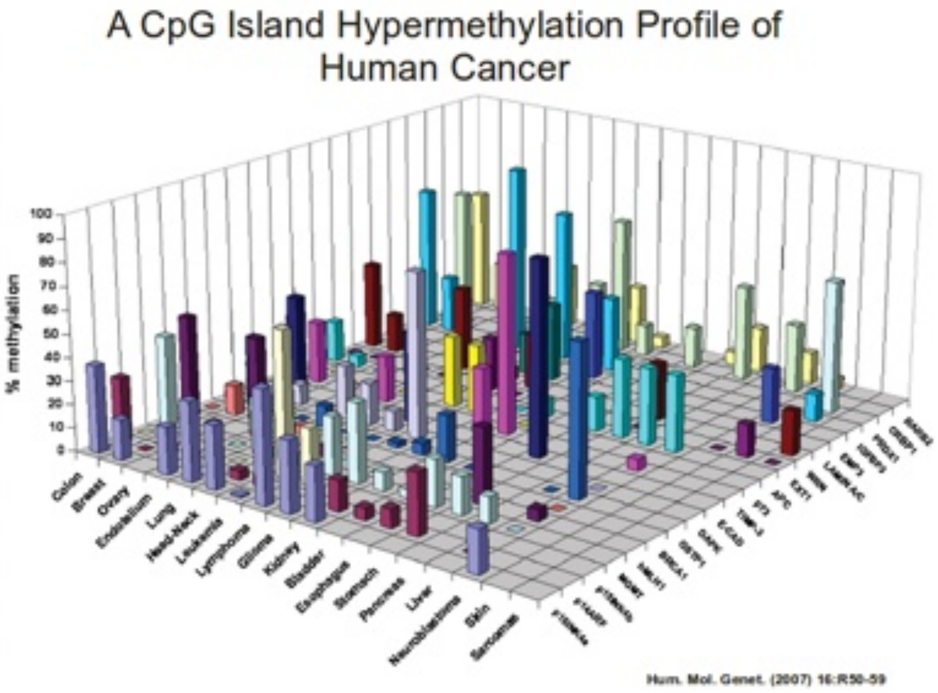


https://gorelik.net/2017/11/23/how-to-make-a-graph-less-readable-rotate-the-text-labels/

3D

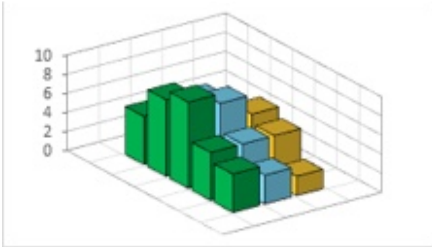
Undvik

Mycket svårt att läsa och jämföra, dolda data

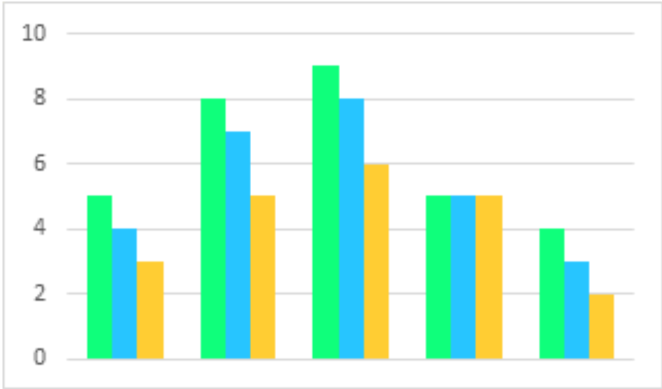


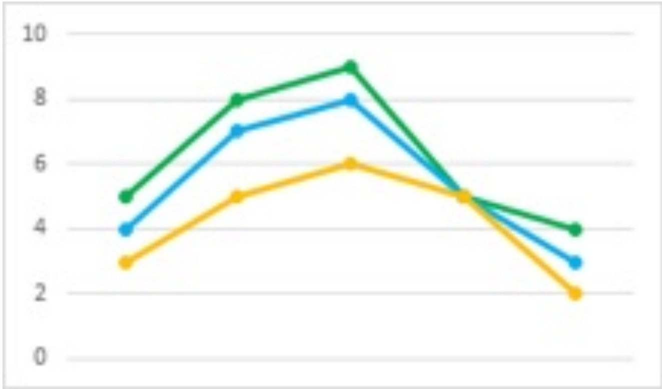
3D

Överväg andra diagramtyper



https://peltiertech.com/3d-bar-chart-alternatives/





CIRKELDIAGRAM

Undvik

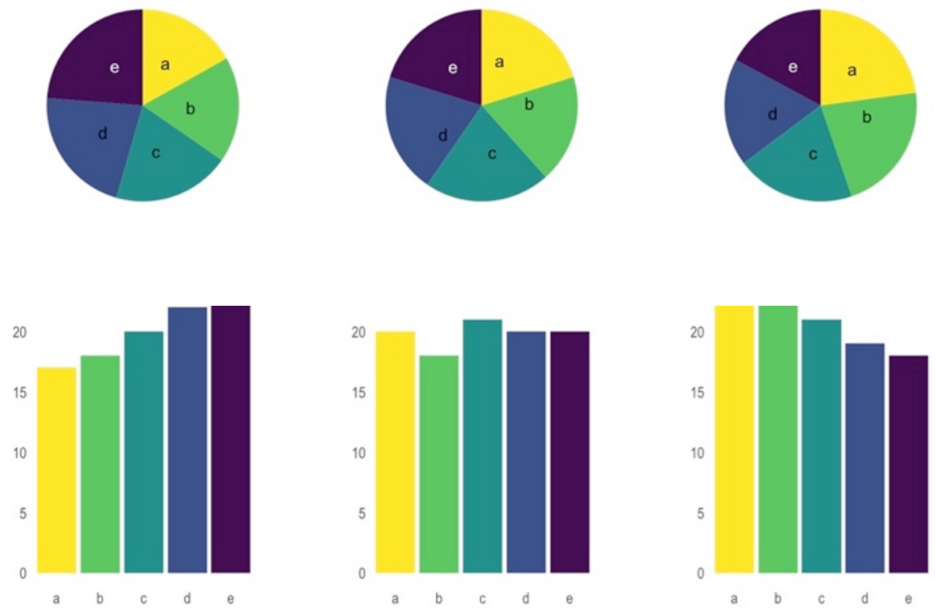
Perceptionsproblem (men inte alltid)

OK för ≤6 kategorier

Kan vara svårt att läsa

CIRKELDIAGRAM

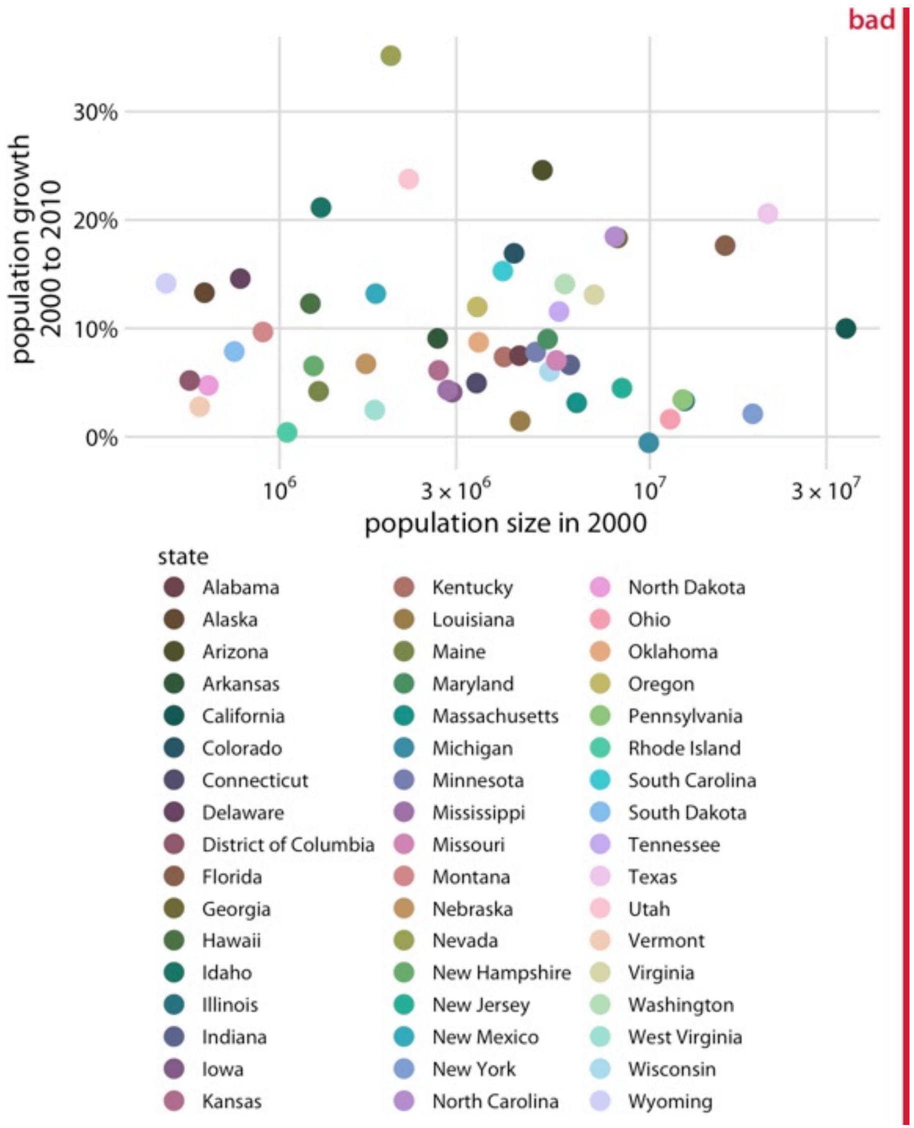
Alternativ: stapeldiagram



https://www.data-to-viz.com/caveat/pie.html

FÖR MÅNGA FÄRGER

OK med ≤ 6 kategorier



https://clauswilke.com/dataviz/color-pitfalls.html

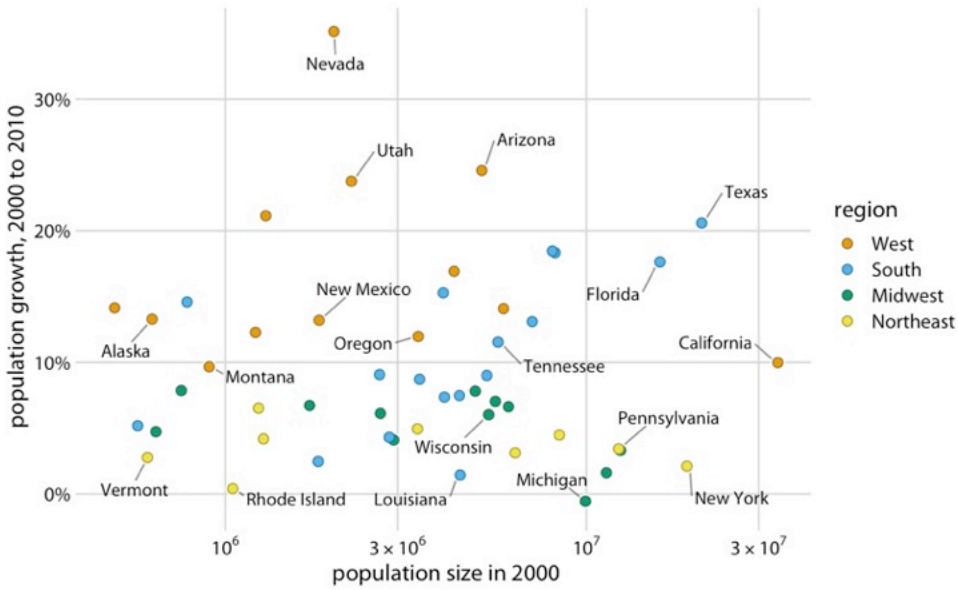
FÖR MÅNGA FÄRGER

Alternativ:

Direkt märkning

Stapeldiagram eller annan diagramtyp

[10 ways to use fewer colors in your data visualizations]

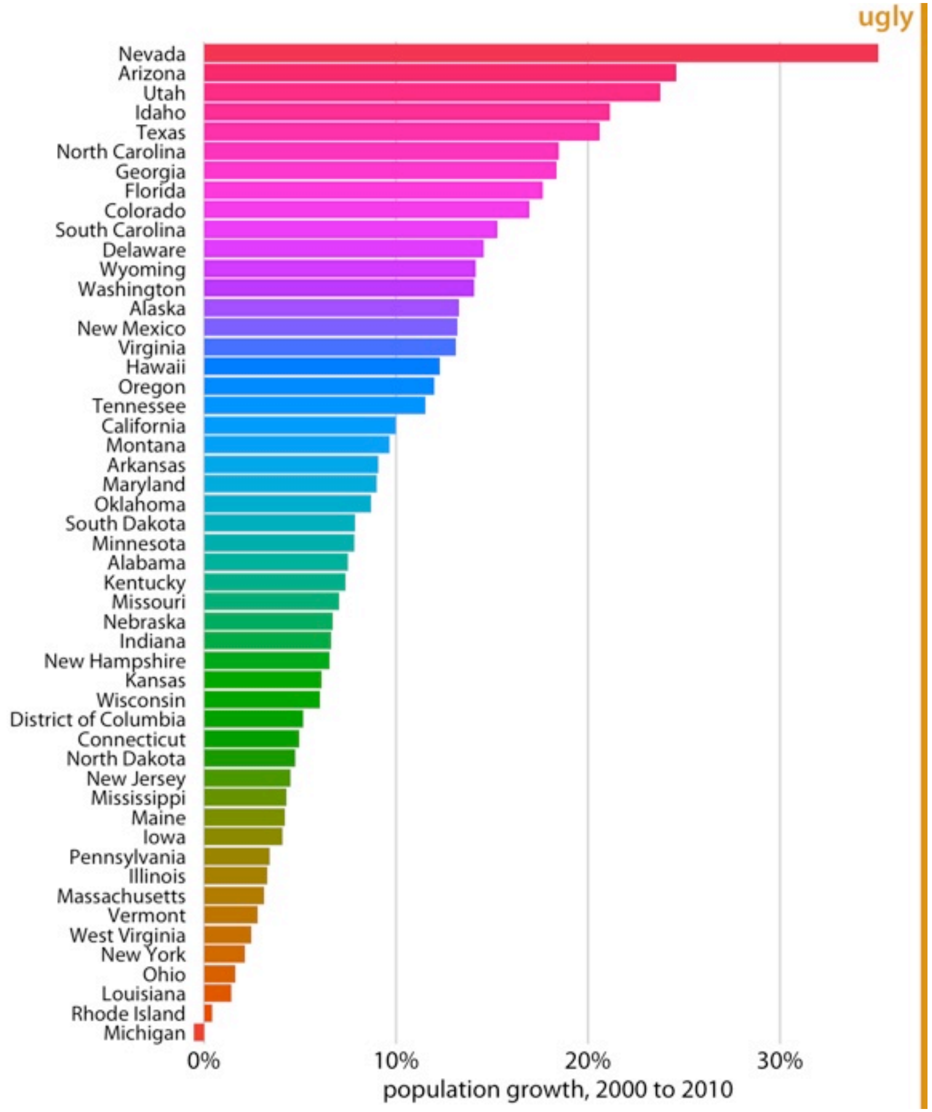


https://clauswilke.com/dataviz/color-pitfalls.html

ONÖDIG FÄRG

När färgen inte tillför någon information

Använd en färg istället



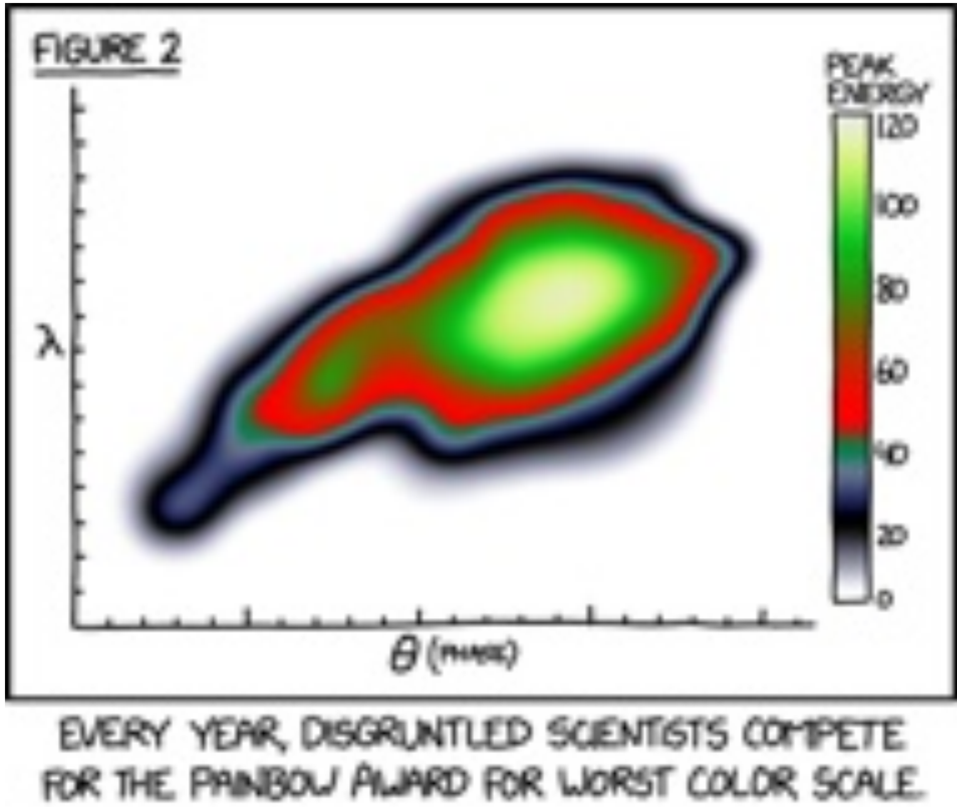
https://clauswilke.com/dataviz/color-pitfalls.html

REGNBÅGSFÄRG-SKALOR

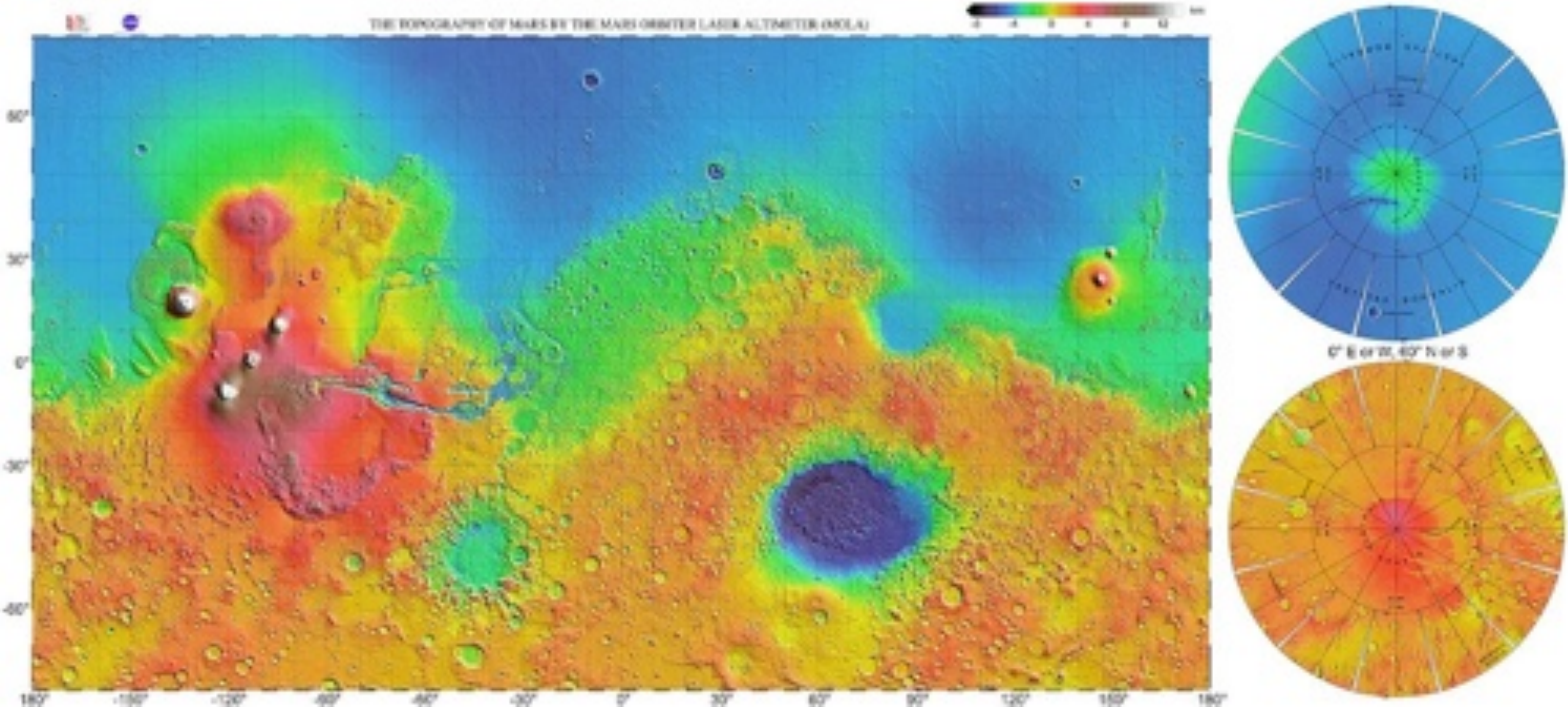
Undvik om möjligt

Var försiktig med standardalternativen i program

[**How rainbow colour maps can distort data and be misleading**]



https://xkcd.com/2537/

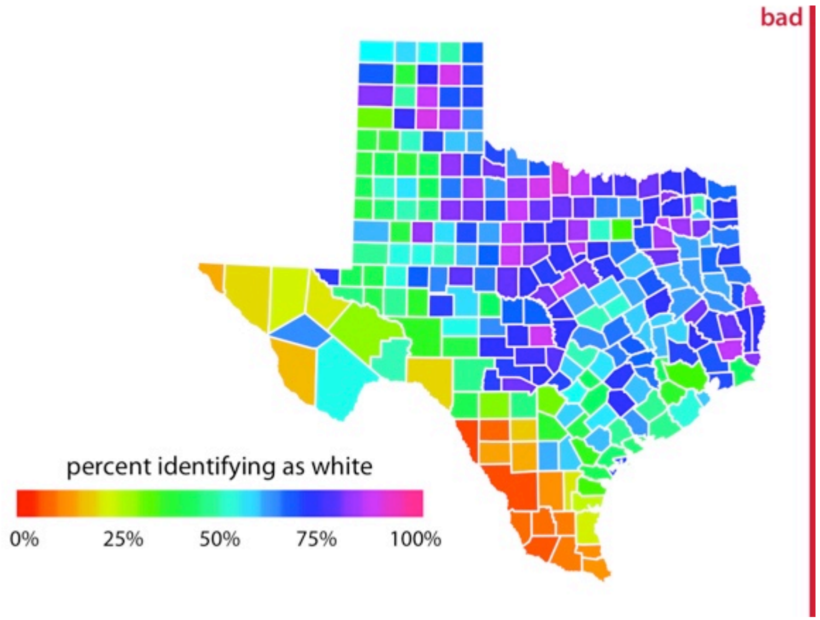


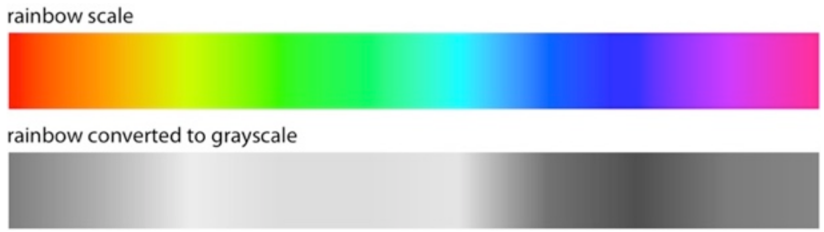
https://attic.gsfc.nasa.gov/mola/images.html

REGNBÅGSFÄRG-SKALOR

Svårt att jämföra värden

Ojämna kulörer och lätthet, missvisande

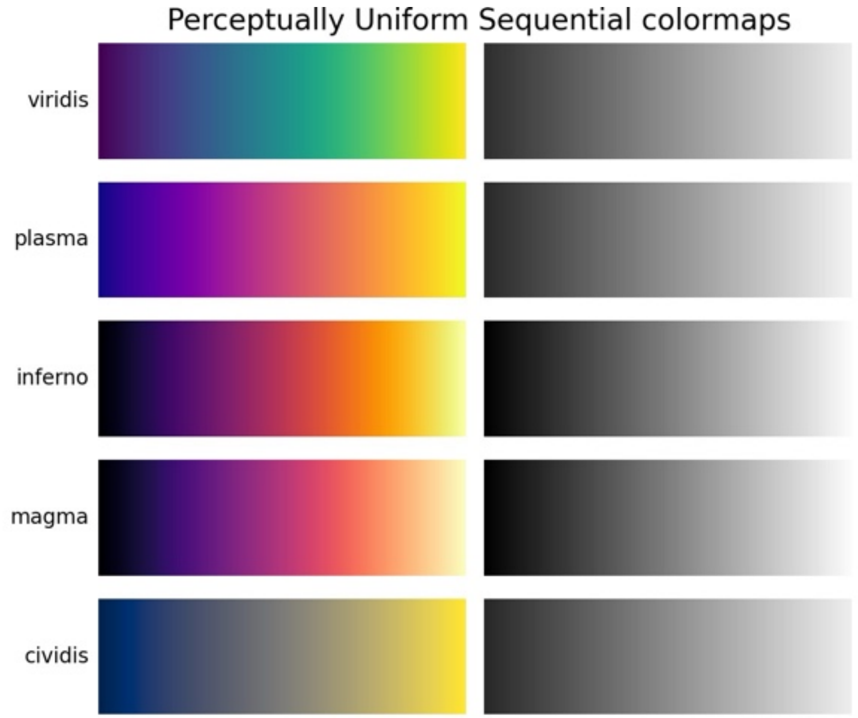




https://clauswilke.com/dataviz/color-pitfalls.html

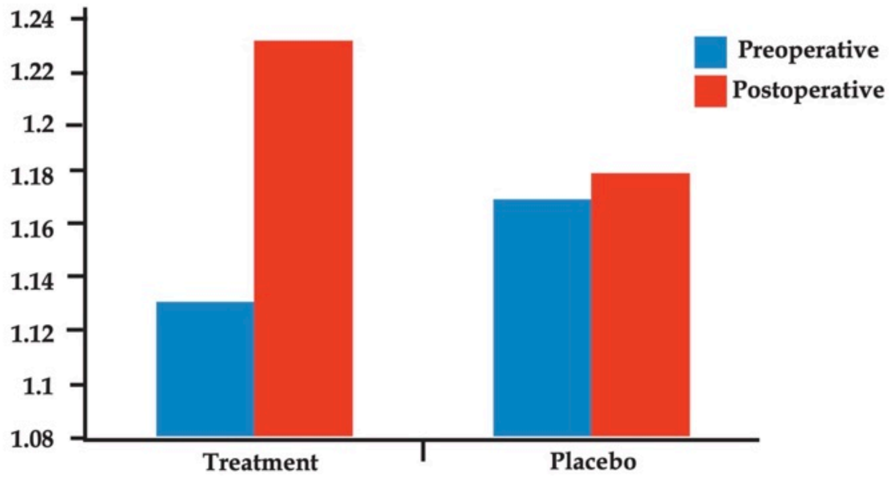
REGNBÅGSFÄRG-SKALOR

Använd enhetliga färgskalor istället



https://matplotlib.org/stable/tutorials/colors/colormaps.html

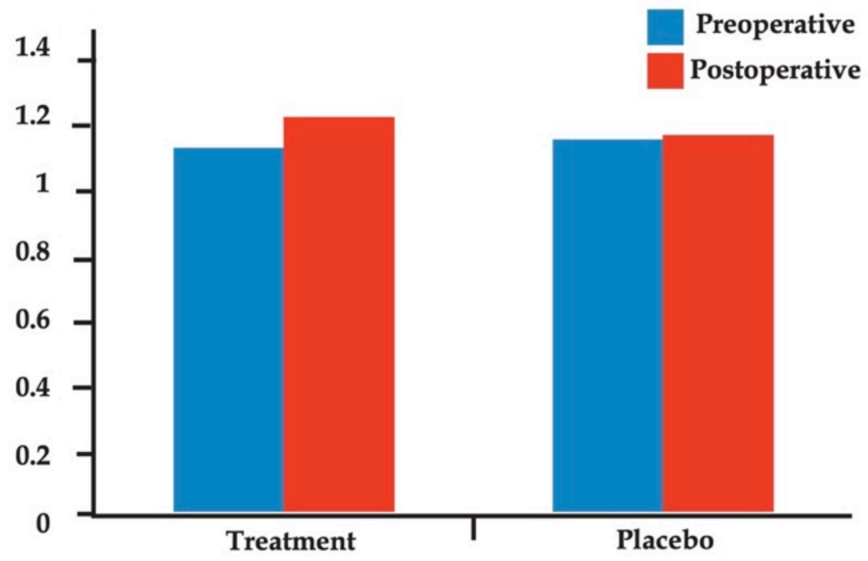
Y-AXELN BÖRJAR INTE PÅ NOLL



Duquia RP, Bastos JL, Bonamigo RR, González-Chica DA, Martínez-Mesa J. Presenting data in tables and charts. An Bras Dermatol. 2014;89(2):280-5

Y-AXELN BÖRJAR INTE PÅ NOLL

I stapeldiagram ska y-axeln börja på noll



Duquia RP, Bastos JL, Bonamigo RR, González-Chica DA, Martínez-Mesa J. Presenting data in tables and charts. An Bras Dermatol. 2014;89(2):280-5

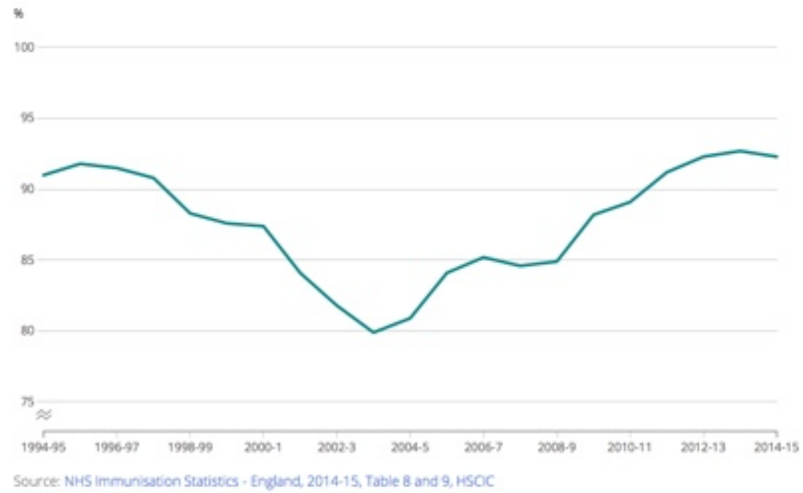
Y-AXELN BÖRJAR INTE PÅ NOLL

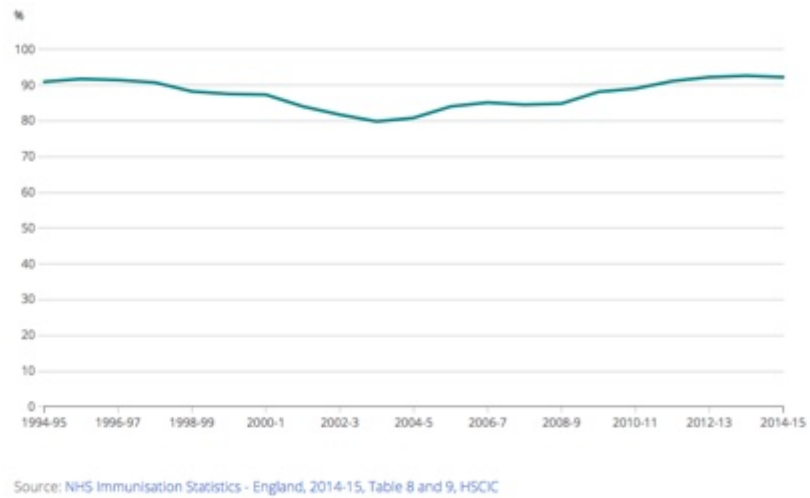
Ingen konsensus om andra diagram

Linjediagram: samma trender men små och viktiga förändringar kan gå förlorade

[**Where to Start and End Your Y-Axis Scale**]()

COMBINED MMR VACCINATION RATE, 1994-5 TO 2014-15, ENGLAND

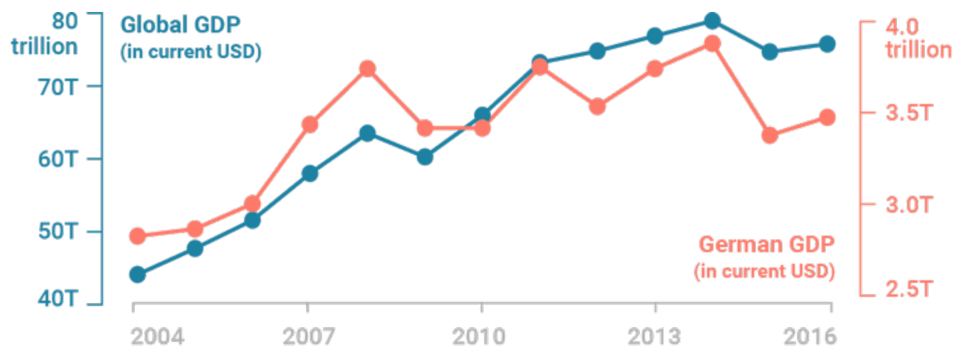




https://digitalblog.ons.gov.uk/2016/06/27/does-the-axis-have-to-start-at-zero-part-1-line-charts/

DUBBEL Y-AXEL

Undvik



https://blog.datawrapper.de/dualaxis/

DUBBEL Y-AXEL

Svårläst

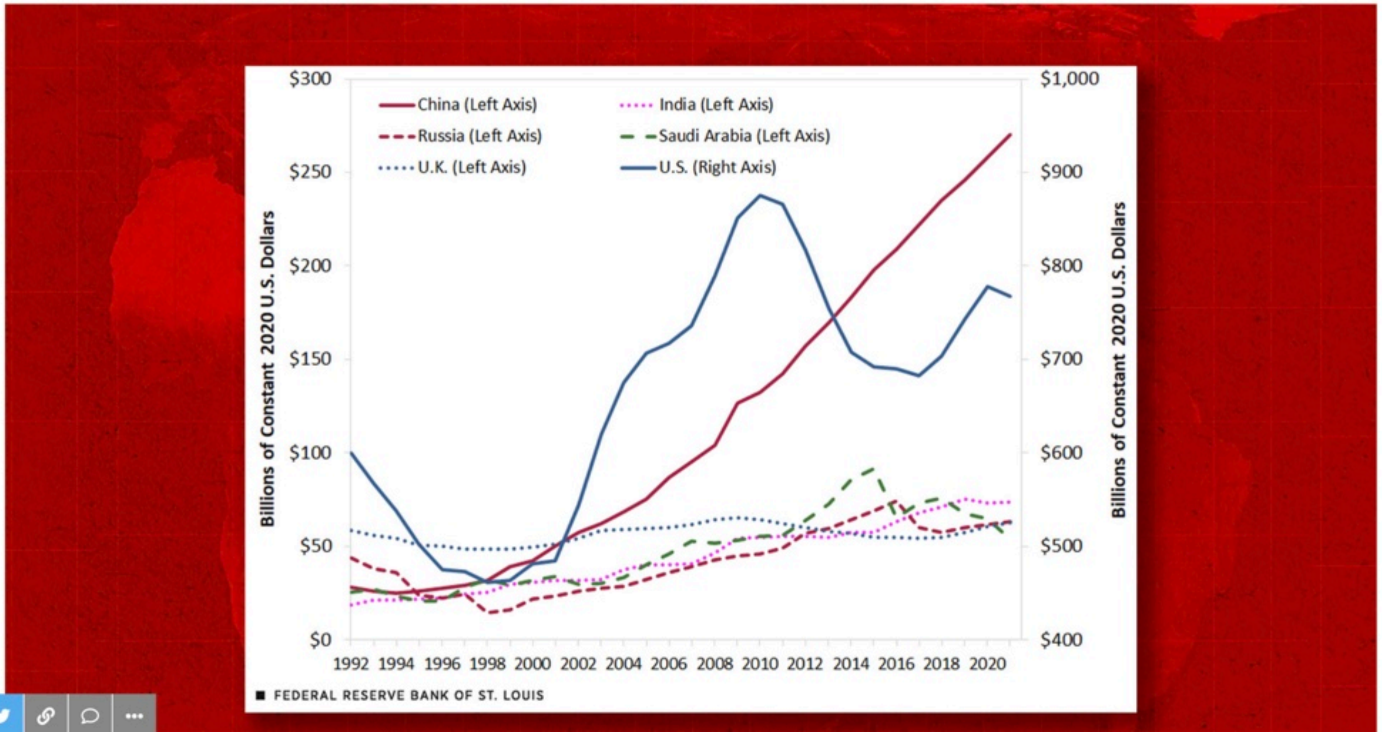
Beroende på axelskalor kan linjerna se väldigt olika ut och i olika positioner

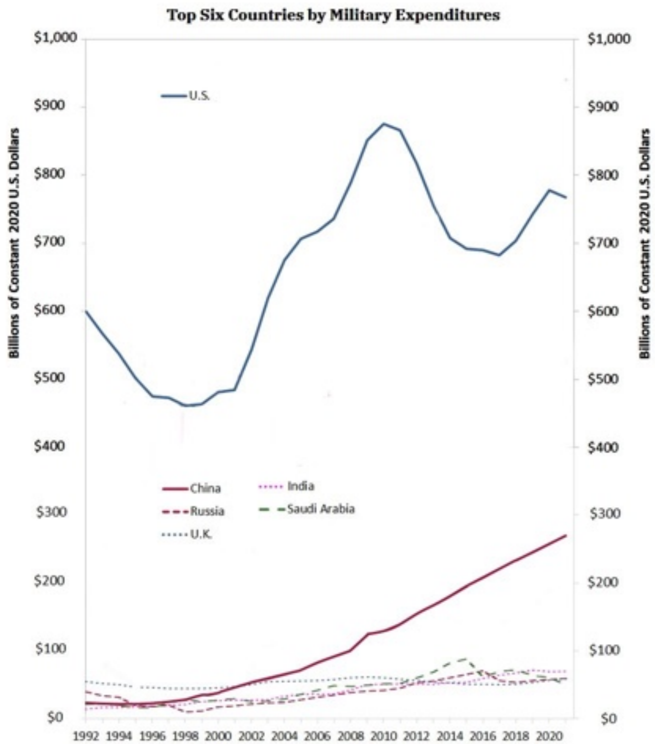
https://blog.datawrapper.de/dualaxis/

Exaggerating China's military spending, St. Louis Fed breaks all statistical rules with misleading graph

The Federal Reserve Bank of St. Louis published a jaw-droppingly misleading graph that portrays China as spending more on its military than the US. In reality, the Pentagon's budget is roughly three times larger.

By Ben Norton Published 2023-01-23





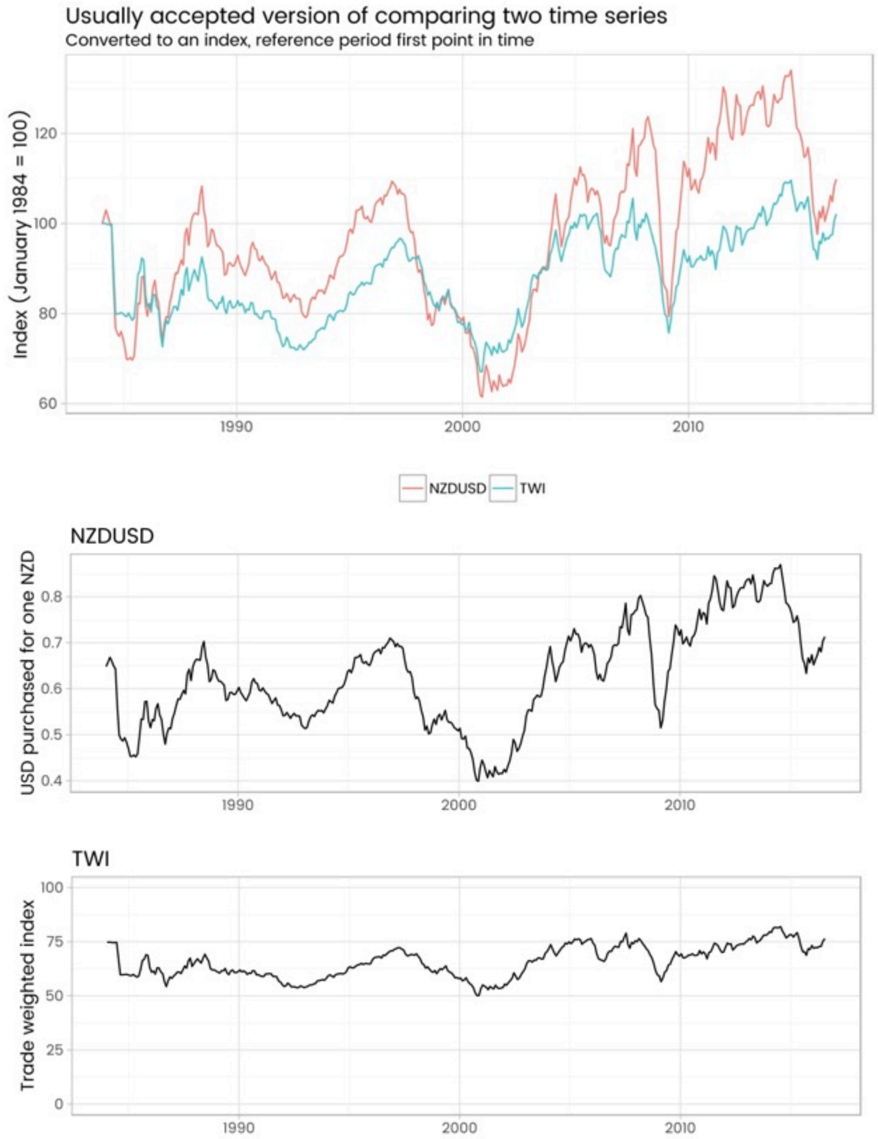
DUBBEL Y-AXEL

Alternativ:

Index

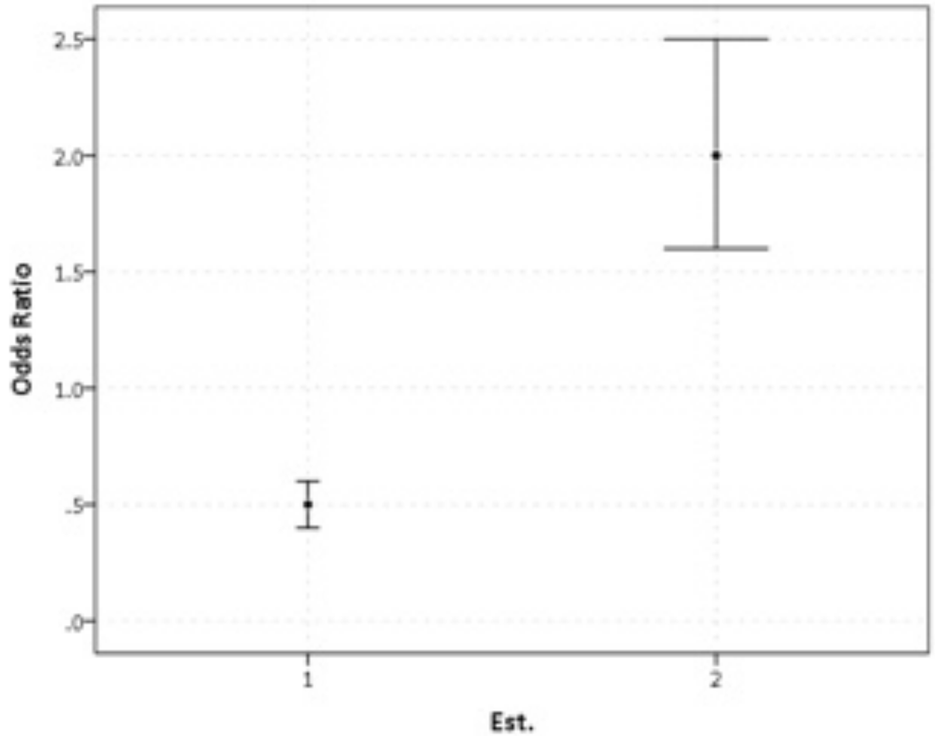
Två diagram bredvid varandra

[**Why not to use two axes, and what to use instead**]()



http://freerangestats.info/blog/2016/08/18/dualaxes

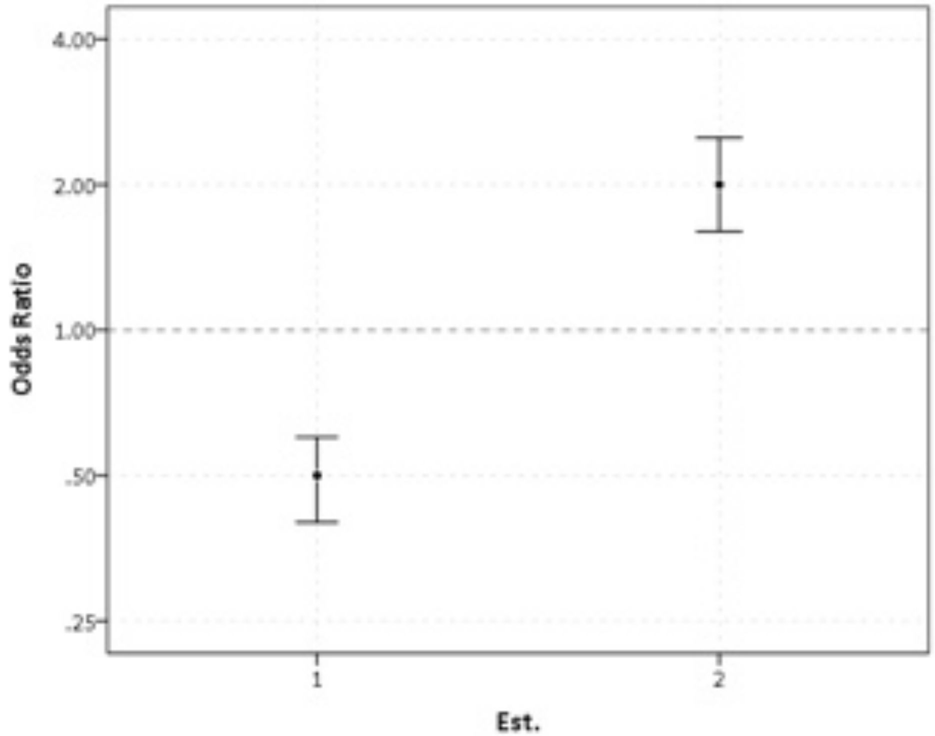
ODDSKVOT UTAN LOGARITMISK SKALA



https://andrewpwheeler.com/2013/10/26/odds-ratios-need-to-be-graphed-on-log-scales/

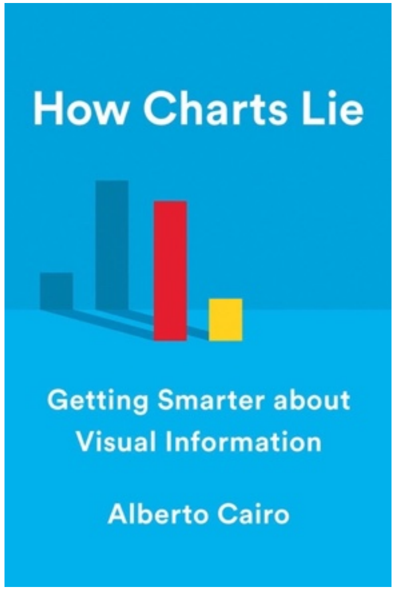
ODDSKVOT UTAN LOGARITMISK SKALA

Använd logaritmisk skala



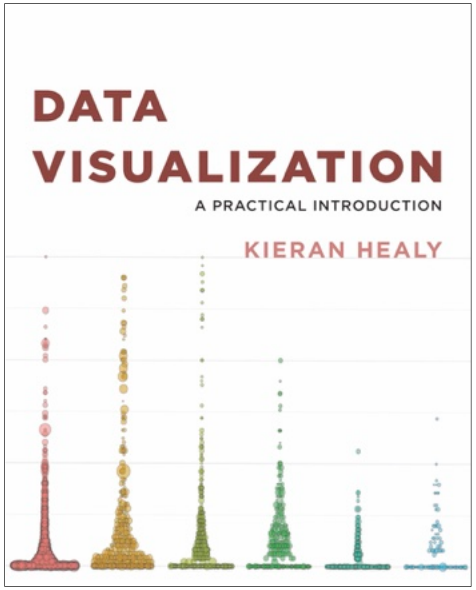
https://andrewpwheeler.com/2013/10/26/odds-ratios-need-to-be-graphed-on-log-scales/

RESURSER



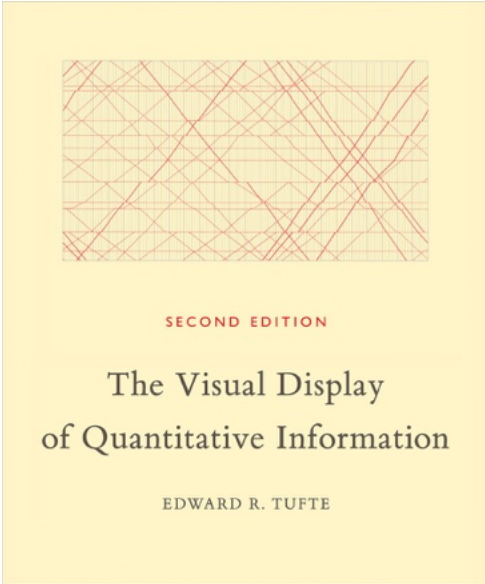
How charts lie: getting smarter about visual information

UU bibliotek



Data visualization: a practical introduction

UU bibliotek
Gratis onlineversion



The Visual Display of Quantitative Information

UU bibliotek

Datawrapper (theory and how-tos, showcase, stories)

Nightingale (tutorials, stories, industry-related news)

FlowingData (showcase)

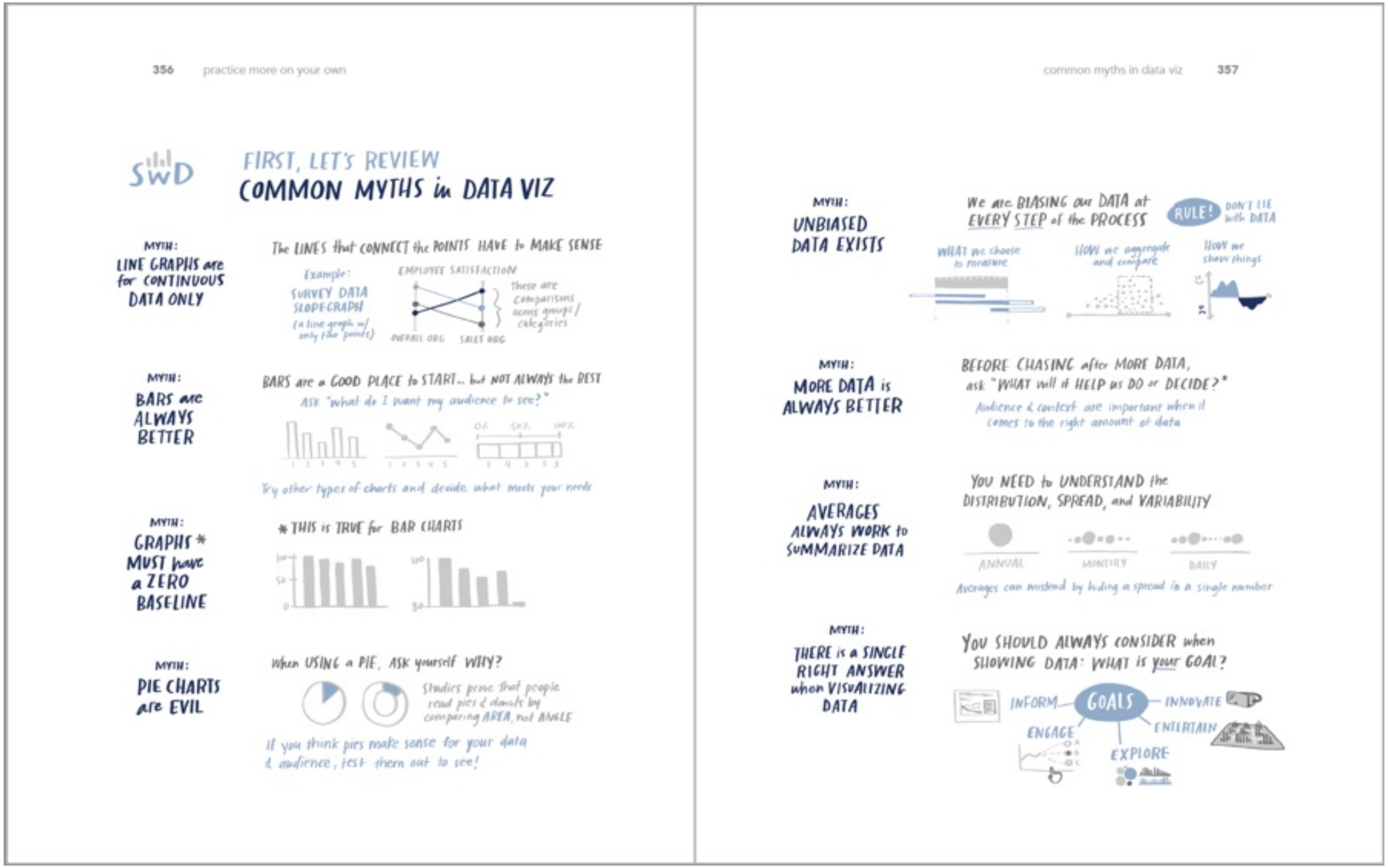
Defense Against Dishonest Charts (interactive examples)

The R Graph Gallery (tutorials with code)

A ggplot2 tutorial for beautiful plotting in R (tutorial with code)

Why scientists need to be better at data visualization

Friends Don't Let Friends Make Bad Graphs



https://www.storytellingwithdata.com/blog/2019/9/30/how-it-came-to-be

MISSVISA INTE!

georgios.karamanis@neuro.uu.se

https://karaman.is

https://github.com/gkaramanis

https://bsky.app/profile/karaman.is

☕ PAUS

GGPLOT2 WORKSHOP

bit.ly/isr-ggplot2

(https://gkaramanis.github.io/ISR-Workshop)